ORACLE: A Persistent Chemical State for Reproducible Molecular Workflows

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Abstract

ORACLE introduces a persistent chemical state for reproducible molecular workflows. Atomic synthons and the effective atomic number combine continuous geometric and electronic evidence into element-preserving local identities that can be compared, clustered and transferred. The state links molecular topology, interactions, symmetry, fragmentation and primitive coordinates with quantum-mechanical observables such as atomic charges, bond orders and spin populations. These observables refine the same chemical representation across structure generation, parameter transfer and optimization, including through overlapping fragments for large molecules. ORACLE also represents reactive contacts in transition states and connects them to downstream coordinate construction. The framework is evaluated across equilibrium and non-equilibrium chemical states, including changes in electronic structure, environment and topology. It is designed to support sophisticated quantum-mechanical studies of chemically relevant molecules of roughly 100–200 atoms in medicinal chemistry, photovoltaics, catalysis and related molecular materials.

1 Introduction

The passage from a molecular geometry to a chemical prediction contains a hidden failure point: the molecule is repeatedly reinterpreted as it moves between file formats, perception libraries, electronic-structure programs, fragment builders and optimizers. A bond, ring, aromatic assignment or reactive contact that changes silently at one boundary can alter every downstream result while leaving each individual program apparently correct. We address this problem by treating perception as a state-transfer problem, not as a preprocessing step. Four explicit tests make this contract measurable: the transition-state capability gap, the state-blind electronic control, non-obvious high-coordination cases, and held-out structural transfer.

The same structure is used repeatedly for different scientific purposes. It may be converted into a graph, divided into fragments, assigned local types, given symmetry or interaction labels, and passed to a quantum-mechanical calculation or an optimizer. Each operation can change the information that is chemically relevant. A useful representation must therefore preserve identity while allowing its geometric, electronic and environmental properties to evolve.

This need is especially clear for molecules of medicinal, photophysical and catalytic interest, where conjugation, weak interactions, stereochemistry, near-symmetry and complex coordination often coexist. The challenge is not only to recognize a structure once, but to keep the same chemical meaning available when the structure is transferred, refined, fragmented or compared with another electronic state.

The new Operator for Recognition, Atom-typing and Continuous Local Equivalence (ORACLE) addresses this challenge by making the interpreted molecule a persistent scientific object. Atomic synthons collect local geometric and electronic evidence, while the effective atomic number provides a continuous, element-preserving coordinate for comparison and transfer. Discrete chemical objects are accepted from this evidence and remain linked to their origin, dependencies and version. Quantum-mechanical results can consequently refine

the representation, and the refined representation can guide subsequent chemical operations.

The method is intended as a support layer for sophisticated quantum-mechanical studies, not as a replacement for electronic-structure theory or optimization algorithms. For large molecules, the same information can be obtained through chemically meaningful overlapping fragments and reconstructed into one auditable object. This creates a common basis for local metrics, parameter transfer, force-field construction, structural correction, transition-state analysis and non-obvious coordination chemistry.

The framework makes three testable contributions. First, it defines a chemically typed state whose continuous descriptors and discrete objects are serialized together, so that identity can be preserved or deliberately invalidated. Second, it makes that state executable across a heterogeneous workflow: ORACLE owns perception and the coordinate optimizer performs the optimization before the returned geometry is re-perceived. Third, it extends perception to single-geometry transition states, where forming and breaking contacts are chemically meaningful but must not be silently promoted to a new constitutional graph. In that setting, numerical conditioning of competing reactive charts becomes an explicit selection rule. The same state-transfer principle extends beyond a single ground-state geometry: the validation includes open-shell multiplicities, excited-state density changes, single-geometry transition states, and coordination topologies beyond the usual four-coordinate regime. These cases make the electronic and coordination state explicit rather than treating them as downstream annotations. The study therefore tests one focused proposition: a molecular state that is continuously perceived and explicitly versioned should remain chemically meaningful and executable as the downstream application changes. We reuse that state for symmetry, macrocycles, interactions, fragment construction, electronic feedback, transition-state optimization, sparse-reference retrieval and structural improvement. The main text presents the state-transfer argument and its discriminating examples; formal schemas, benchmark tables, software contracts and reproduction commands are provided in the Supporting Information. Figure 1 summarizes the scope of the framework; the termi-

Table 1: Terminology used in the main text. Software and archive details are given in the Supporting Information.

Term	Meaning in this work
ORACLE	Persistent chemical-state and perception layer.
SMILES Writing and Interpretation of Topology, Chirality and Hydrogens (SWITCH)	Production chemoinformatics layer replacing RDKit in the workflow.
Symmetry and coordinates	Symmetry and primitive-coordinate construction / resulting coordinate set.
Coordinate optimization	Internal-coordinate optimization using the persistent chemical state.
L0, L1, PL1	Baseline electronic state, higher-level geometry, and posterior-corrected geometry.
Reference libraries	Published and extended structure libraries used for structural transfer.
General Amber force field, version 2 (GAFF2)	External atom-type vocabulary used only for interoperability.

nology, persistent-state distinction and capability inventory are listed in Tables 1, 2 and 3, respectively.

2 The Molecular State as the Unit of Continuity

The comparison is deliberately functional. Cheminformatics toolkits provide graph construction, file conversion and substructure operations; workflow engines provide execution and provenance; electronic-structure interfaces provide access to energies and densities. ORACLE adds a chemical-state contract that connects these layers, keeping continuous evidence, accepted objects, electronic metadata, dependency invalidation and provenance attached to the same molecular identity.

Within the chain outlined above, the method treats perception as a persistent chemical state that remains invariant until a declared dependency changes. This scientific contract distinguishes it from application-local perception: continuous evidence, accepted discrete objects, provenance and dependencies travel together as one molecular object.

The integration problem is to move from minimal input to mutually consistent bond perception, aromaticity, symmetry, atom typing and fragmentation without allowing each application to redefine the molecule. The present workflow returns continuous chemical

descriptors and atom types, paired charges and bond orders, rings and aromaticity, exact and local symmetry, fragments and, when requested, constructed or optimized structures.

For transition-metal and organometallic systems, an enriched Cartesian record is the primary representation. It carries molecular charge, multiplicity, accepted connectivity, coordination edges, hapticity and metal–metal relationships explicitly, providing a natural description of delocalized and multicenter coordination. A quantum-chemistry schema (QCSchema)-compatible molecule record supplies an interoperable outer container, while the enriched XYZ state retains the perception audit. Extended SMILES (CXSMILES) with directed dative bonds provides an optional exchange format, and the enriched record preserves the authoritative chemical state. When Cartesian coordinates are unavailable, a declarative coordination specification records the metal centers, molecular charge and multiplicity, ligand inputs and donor atoms, denticity, hapticity, bridges, metal–metal bonds and coordination-geometry hypotheses. Template and fragment assembly then generates auditable Cartesian candidates. Explicit enumeration preserves alternative oxidation and spin states through the scoring stage. The user interface remains automatic, while every threshold, residual and source is available for inspection.

Existing cheminformatics, symmetry, molecular-editor and optimizer packages already provide many of the required individual operations.[1–6] The contribution is the chemical boundary that allows those operations to cooperate. Continuous coordination, bond-order and local-environment evidence first stabilizes perception; accepted rings, symmetry operations, chemical descriptors and fragments then become exact, versioned objects. Optimized graph, coordinate and numerical machinery, including compiled kernels and an internal electronic-structure interface, supports unattended execution while preserving the same audit contract.

This scope places ORACLE squarely in computational-chemistry methodology: the persistent state is the substrate for efficient geometric and electronic metrics, chemistry-guided overlapping fragmentations, state-specific atom typing, and accurate force-field parameter

transfer. The effective atomic number follows the broader atoms-in-molecules tradition of representing a quantum-mechanical environment through local atomic objects, while extending that idea with continuous geometry, coordination, charge, bond-order and spin evidence that remains executable across workflows.[7] ORACLE therefore connects quantum-mechanical (QM) interpretation to practical molecular modelling as a chemical state layer for reproducible workflows. The demonstrated scale is already relevant to this use: paclitaxel and vancomycin benchmark the large-molecule perception path, while [6]helicene and the 176-atom ablation member show state transfer and overlap-aware processing beyond small-molecule examples.

The scope is complementary to established workflow frameworks. Fragment assembly and quantum-mechanical (QM) fragmentation are established routes for extending high-level electronic information to larger molecular systems, while restrained electrostatic potential (RESP)-type charge models illustrate the importance of a reproducible QM-to-parameter interface.[8,9] Open Babel and RDKit provide mature graph and cheminformatics operations; ASE provides a general atomistic calculator and structure-management layer; and Psi4 and QCArchive/QCEngine-style infrastructures organize quantum-chemical jobs and their results.[1,2,10,11] ORACLE addresses a different object: the chemically interpreted state that links topology, symmetry, fragments, interactions and primitive coordinates before those workflow systems consume or return a calculation. This distinction defines the present contribution as workflow methodology with a chemical state contract, while the established frameworks remain appropriate execution and interoperability layers.

This persistent chemical layer complements cheminformatics, geometry optimizers, force fields and electronic-structure software. It lets all components operate on the same versioned molecular object. The chemical representation module is shown in Figure 2.

Its distinguishing feature is the conjunction

$$\text{continuous evidence} \;+\; \text{audited discrete acceptance} \;+\; \text{persistent reuse}. \tag{1}$$

Table 2: Application-local perception compared with the persistent chemical state.

Aspect	**Application-local perception**	**Persistent layer**
Lifecycle	Reconstructed for the current application	Perceived once and reused through explicit updates
Representation	Primarily discrete molecular graphs	Coupled continuous evidence and accepted discrete objects
Provenance	Recorded at the level selected by each application	Stored with thresholds, residuals and data sources
Dependencies	Managed independently by individual consumers	Versioned dependencies and targeted invalidation
Chemical coverage	Unsupported environments may be handled locally	Explicit missing-coverage diagnostics and reference requests

Table 3: Chemical capabilities provided by the ORACLE representation.

Object or operation	**Role in ORACLE**
Atomic synthon	Continuous local identity combining geometry, coordination, charge, bond-order and spin evidence.
Effective atomic number	Element-preserving continuous coordinate for comparing, clustering and transferring local environments.
Discrete chemical state	Audited graphs, rings, interactions, symmetry operations and primitive coordinates accepted from continuous evidence.
Reactive perception	Forming/breaking contacts and alternative primitive charts for transition-state geometries; chart selection uses the normalized Wilson-B condition number.
Electronic-state awareness	Complete Charge Model 5 (CM5), Mayer and spin-resolved descriptors update the same state for open-shell, excited-state and solvent comparisons.
Extended coordination	The same representation handles fragment assemblies and non-obvious metal environments, including $Fe_4S_4Cl_4$.

Continuity alone permits incompatible downstream graphs, whereas persistence alone can freeze a brittle cutoff; the framework therefore couples the two.

The scientific contribution lies in this coupling of continuous perception, explicit discrete acceptance and persistence. Its central chemical module is the atomic synthon together with the effective atomic number. Here, an *atomic synthon* is a local, element-preserving descriptor that collects the geometry, coordination and quantum-mechanical observables of one atom together with the bond-order information incident on it. The continuous environmental term ET_i summarizes this local evidence for atom i; the effective atomic number is defined as $Z_i^{\mathrm{eff}} = Z_i - \frac{1}{2} + \mathrm{ET}_i$, where Z_i is the ordinary atomic number. The synthon collects local geometric and electronic evidence, while Z_i^{eff} supplies a compact elemental–environment coordinate. These objects make local environments comparable, element-preserving, clusterable, and transferable across geometries and electronic states. Established Horton cycles, Kabsch alignment, molecular mechanics and electronic population models instantiate bounded operations within that contract.[12–14] Accepted objects remain invariant until their dependencies change; targeted invalidation removes stale identities; and overlap constructs larger states from validated fragments. Chemically irrelevant perturbations therefore preserve identity, exact symmetries and cycles remain executable, and different calculations retain a consistent molecular interpretation. Serialization, provenance, dependency tracking and compiled kernels make this model robust, auditable and usable.

3 Constructing an Executable Molecular State

3.1 The perception–calculation boundary

From simplified molecular-input line-entry system (SMILES) or an enriched Cartesian/QCSchema record, the workflow builds an inspectable molecular state with immutable snapshots. Here "immutable" refers to a serialized snapshot: it is never edited in place. A geometry, electronic result, accepted graph, or downstream structural operation that changes a declared

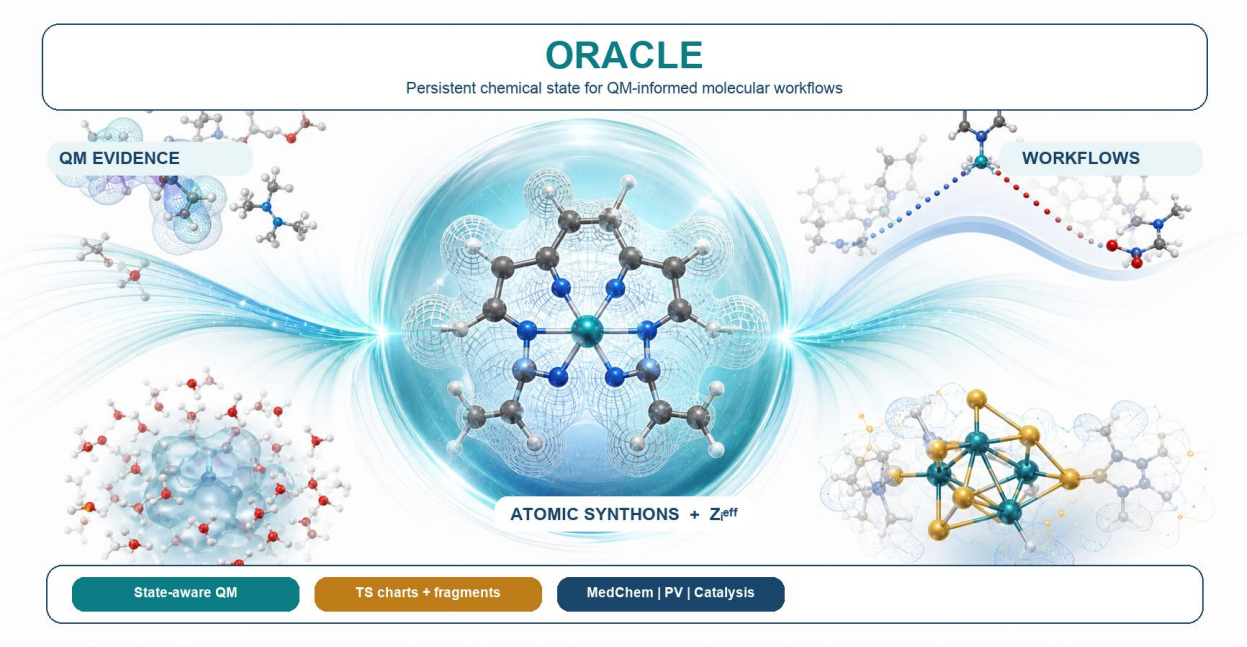


Figure 1: Persistent chemical state connecting quantum-mechanical evidence with molecular workflows.

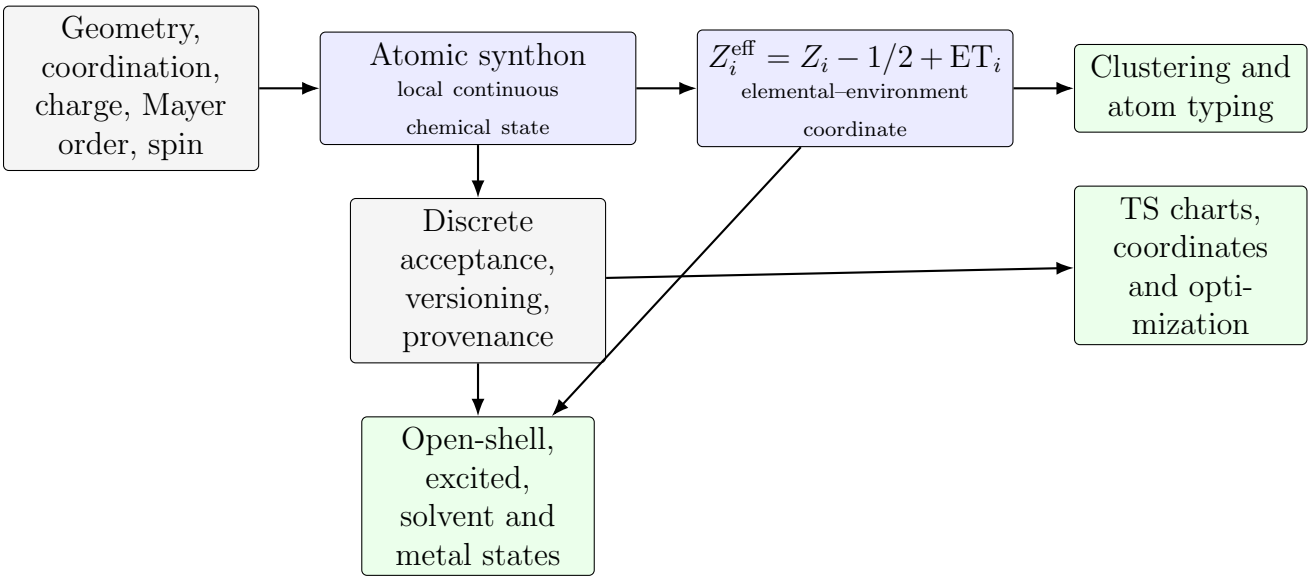


Figure 2: Continuous chemical evidence organized into versioned molecular objects.

dependency creates a new state version with a parent fingerprint; unchanged dependencies permit reuse of the previous snapshot. Thus "perceived once" means once per version, not once for all future geometries, and "persistent" means persisted and reused until targeted invalidation is triggered. (Figure 3). Its primary chemical products are atom types and local chemical descriptors (hereafter synthons);[15] paired charges and bond orders; canonical rings and aromaticity; exact symmetry operations and equivalence classes; hydrogen-bond contacts; and alternative fragment views. When requested, the same state drives overlap reconstruction or an explicit geometry-optimization route. Graphs, primitive coordinates and analytic Wilson matrices are generated internally to support these products and remain available for audit.

Four layers remain distinct throughout the workflow: quantum-mechanical evidence; the persistent molecular state that stores it; discrete objects accepted from that evidence; and downstream operations that consume those objects. This separation makes it possible to update charges, bond orders, spin or solvent metadata without silently replacing the molecular identity, while allowing a changed dependency to trigger a controlled new state.

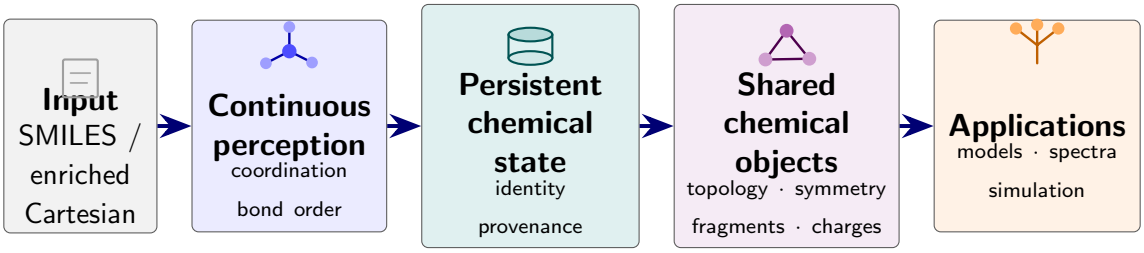


Figure 3: A persistent perceived state shared by downstream applications.

Black-box operation combines automatic selection of ring algorithms, symmetry frames, fragment overlaps, electronic-population analysis and numerical backends with inspectable evidence and provenance. Applications transform the stored state through explicit, versioned updates.

For clarity, the terminology follows three levels. *Continuous evidence* denotes coordination, charge, bond order and the derived local descriptors. The *persistent state* contains the exact graph, rings, symmetry, atom classes and fragments accepted from that evidence.

Named service programs are defined together in the Computational Details. They supply input operations and electronic observables without changing this chemical hierarchy.

3.2 Continuous evidence and discrete chemical objects

The descriptor construction follows three chemically intuitive questions. First, how many neighboring atoms influence a center, and with what smoothly varying strength? Second, how are charge and bond order distributed through that environment? Third, which local environments remain equivalent when geometry, coordination or electronic populations change slightly? The continuous coordination number answers the first question with a smooth neighbor count. The atomic synthon gathers the geometric and electronic answers into a local descriptor, while the effective atomic number provides a single continuous coordinate for comparing related environments. These quantities connect smoothly varying chemical evidence to elemental identity and the accepted exact graph.

The continuous coordination number (CNA) illustrates the evidential layer,

$$\mathrm{CNA}_i = \sum_{j \neq i} \frac{1}{2} \left[1 + \mathrm{erf}\big(\alpha_{\mathrm{CNA}} (R^0_{ij} - R_{ij}) \big) \right], \tag{2}$$

with coordination-dependent covalent radii obtained by smooth Hermite interpolation of Pyykkö data.[16] The resulting connectivity weight supplies topological evidence; the stored chemical bond order remains the single value used by downstream models.

To prevent a similar ambiguity in the electronic description, the state stores one charge and one bond order at a single, internally consistent availability level. Complete Charge Model 5 (CM5) charges and Mayer bond orders define the quantum-mechanical (QM) level for the whole structure;[17,18] otherwise the state adopts the paired electronegativity/Pauling geometry estimate. The selected bond order is resolved continuously into non-negative σ, π and $\pi\pi$ components whose sum recovers the original value.

The local descriptors used in this workflow are defined as follows. With $\mathcal{N}(i)$ the contin-

uously perceived neighbors, $\widetilde{R}^{\mathrm{cov}}$ the coordination-adjusted covalent radius and $\bar{R}_i$ the mean incident distance, the compactness and radial-dispersion components are

$$C_i = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \frac{\widetilde{R}_i^{\mathrm{cov}} + \widetilde{R}_j^{\mathrm{cov}}}{R_{ij} + \widetilde{R}_i^{\mathrm{cov}} + \widetilde{R}_j^{\mathrm{cov}}}, \tag{3}$$

$$D_i = \frac{1}{|\mathcal{N}(i)|\bar{R}_i} \sum_{j \in \mathcal{N}(i)} |R_{ij} - \bar{R}_i|. \tag{4}$$

The third geometric component measures angular strain and extends the same description beyond ordinary organic valence. The method forms a continuous electronic-domain (ED) count N_i^{ED} from coordination, lone-pair and open-shell domains and evaluates

$$S_i = \left[\frac{1}{\binom{N_i^{\mathrm{ED}}}{2}} \sum_{j<k} \left(\cos\theta_{ijk} - \cos\bar{\theta}(N_i^{\mathrm{ED}}) \right)^2 \right]^{1/2}. \tag{5}$$

The reference mean angles for domain counts 2, 3, 4, 5, 6, 8 and 12 are 180.0°, 120.0°, 109.47°, 120.0°, 90.0°, 90.0° and 63.43°, respectively, with smooth Hermite interpolation between integer counts. Equation 5 therefore treats tetrahedral, hypervalent and high-coordinate environments within one expression.

These components, the atomic charge and the incident bond-order sums define the seven-component atomic synthon

$$\mathbf{s}_i = (q_i, C_i, D_i, S_i, I_i^{(\sigma)}, I_i^{(\pi)}, I_i^{(\pi\pi)}), \tag{6}$$

where $I_i^{(x)} = \sum_{j \in \mathcal{N}(i)} \mathrm{BO}_{ij}^{(x)}$, $x \in \{\sigma, \pi, \pi\pi\}$, collects each non-negative incident bond-order component. These definitions remain continuous in geometry, coordination and bond order.

Higher-order synthons are built compositionally:

$$\mathcal{S}_{ij}^{(\mathrm{bond})} = (\mathbf{s}_i, \mathbf{s}_j, \mathrm{BO}_{ij}) \,, \tag{7}$$

$$\mathcal{S}_{ijk}^{(\mathrm{angle})} = \left(\mathcal{S}_{ij}^{(\mathrm{bond})}, \mathcal{S}_{jk}^{(\mathrm{bond})}, \cos\theta_{ijk}\right) , \tag{8}$$

$$\mathcal{S}_{ijkl}^{(\mathrm{dihedral})} = \left(\mathcal{S}_{ijk}^{(\mathrm{angle})}, \mathcal{S}_{jkl}^{(\mathrm{angle})}, \cos\phi_{ijkl}, \cos^2\phi_{ijkl}\right) . \tag{9}$$

Ring synthons collect the ordered bond and angle descriptors of a canonical cycle; hydrogen-bond synthons add the donor–hydrogen–acceptor descriptors and continuous contact geometry. The same effective-atomic-number coordinate combines elemental identity with the local environment. Complete normalizations and similarity metrics are given in the Supporting Information.

Atomic types are obtained by element-preserving complete-link clustering of the atomic synthons, with every accepted class retaining its centroid, maximum descriptor spread and generating thresholds. To connect this native description with an established force-field vocabulary, the workflow also translates the accepted graph, aromatic annotation and bond-order components into General Amber Force Field, version 2 (GAFF2), labels.[19] For saccharin, this maps aromatic carbon, carbonyl carbon, sulfur-bound oxygen, amide nitrogen and hypervalent sulfur to `ca`, `c`, `o5`, `n` and `s6`, respectively; the carbonyl oxygen is represented by the distinct `o` class. The versioned mapping leaves the continuous native type unchanged and records mapping coverage explicitly; complete rules, including phosphorus and sulfur, are given in the Supporting Information.

Synthon-derived classes preserve continuous chemical similarity for equivalence, fragment transfer and model construction. GAFF2 labels connect the same perceived state to an established force-field vocabulary. The saccharin labels below document this interoperable path while every perception decision remains anchored to the underlying continuous evidence.

Equation 6 and its weighted sums admit unrestricted coordination and therefore avoid the tetravalence ceiling implicit in many organic-chemistry typing schemes. The same repre-

sentation consequently describes conventional organic environments and highly coordinated centers. In fixed FeF_n coordination spheres with $n = 4, 6, 8, 10$, and 12, the accepted central degrees are exactly n and the corresponding continuous coordination numbers are 3.99, 5.98, 7.97, 9.97, and 11.96. Radial expansion of the 12-coordinate sphere gives a smooth decay from approximately 12 to zero across a single coordination-independent expression. The perception layer and its covalent-radius support span elements 1–118. Quantitative electronic descriptors use a complete internally generated or user-supplied charge/bond-order pair.

Continuous evidence leads to exact topology. Once a covalent graph has been accepted, deterministic Horton candidates, recursive chord splitting and binary independence selection produce a canonical minimum-length cycle basis of rank $|E| - |V| + c$.[12,20,21] Each cycle is serialized as an ordered canonical atom sequence. Ring systems, atom–ring and bond–ring incidence, and label-preserving automorphisms are then exact objects. Aromaticity is stored separately as a conservative annotation accepted from continuous delocalization evidence together with the local graph and electronic environment.

Hydrogen bonds make this continuous–discrete distinction especially clear. Smooth distance and angular terms, referenced to van der Waals radii, quantify the evidence for a donor–H$\cdots$acceptor contact,[22] while an accepted contact remains distinct from the covalent edge set. An intramolecular contact may therefore close a pseudocycle through the covalent path, whereas an intermolecular contact joins otherwise separate covalent components only at the interaction resolution. The current implementation recognizes chemically compatible nitrogen, oxygen, phosphorus and sulfur sites, and the quantitative response library described below extends the parent hydride environments with systematically substituted monomer/dimer combinations.

A distinct, geometry-certified shared-hydrogen motif is resolved before the ordinary one-neighbor hydrogen constraint is enforced. For each hydrogen, candidate B, N, O, P, and S neighbors must lie within the sum of the corresponding covalent radii plus 0.60 Å. The two nearest candidates are accepted only when their distances agree within 0.15 Å or 10%,

and when the heavy-atom–H–heavy-atom angle is at least 140° for N/O/P/S bridges or 60° for a boron-containing three-center bridge. A certified hydrogen is then allowed exactly the two detected neighbors; every other hydrogen remains monovalent. The two contacts enter the persistent graph, cycle inventory, and provenance together, which covers symmetric proton-transfer structures and B–H–B bridges without molecule-specific templates.

3.3 Electronic closure at the L0 level

To close the perceived state at the level required by its electronic descriptors, the workflow first retains any complete, provenance-bearing set of charges and bond orders supplied by the user. When such data are absent, it generates them with the new L0 baseline: a gas-phase calculation using the parameter-free Perdew–Burke–Ernzerhof hybrid with 25% exact exchange (PBE0) and the Karlsruhe valence triple-zeta basis with polarization (def2-TZVP).[23,24] This parameter-free combination provides a consistent operational baseline for populations and structures over the element range supported by the corresponding all-electron bases and effective-core potentials.

The calculation is performed through an *internal* in-memory interface to the Python-based Simulations of Chemistry Framework (PySCF), which supplies Complete Charge Model 5 (CM5) charges and Mayer bond orders from the same electronic state.[17,18,25] Calculations use tight self-consistent-field convergence and the native scalar-relativistic def2 effective-core potentials from rubidium through radon, which define the current L0 electronic domain. Complete electronic arrays select L0; otherwise the paired electronegativity/Pauling estimate applies to the whole object. Settings and inputs are reported in the Supporting Information. L0 is a single fixed-geometry closure; for larger molecules its overlapping-core and halo construction distributes the QM work across connected fragments.

For a large molecule, the same electronic closure is obtained from overlapping local calculations. Every atom belongs to exactly one fragment core, every perceived covalent pair occurs in at least one L0 core-plus-halo fragment, and the fragments together form a con-

nected overlap graph. Core observations reconstruct the molecular charges, while repeated halo observations quantify their sensitivity to the artificial fragment boundary. Once these local values have been assembled, the unique minimum-weight displacement restores the molecular charge,

$$q_i = q_i^{\mathrm{raw}} - \frac{Q_{\mathrm{raw}} - Q_{\mathrm{mol}}}{w_i \sum_j w_j^{-1}}. \tag{10}$$

Mayer values come from the fragment in which each pair lies deepest, averaging only observations of equal interior rank. Overlap dispersion remains an atomwise boundary diagnostic. Core, bond, overlap-graph and electronic-level completeness are verified before assembly; neutral electrostatic groups for fast-multipole evaluation are constructed afterward. The default two-bond halo expands whenever a boundary would cut a conjugated, ring, metal-coordination or otherwise protected environment.

The frozen Gly–Ala–Gly audit gives a concrete large-state example: three core-plus-two-shell fragments cover all 27 atoms and all 26 covalent bonds, with one PBE0/def2-TZVP single point per fragment. Relative to the direct full-molecule calculation, the assembled CM5 vector has a root-mean-square (RMS) error of 0.0032 e and a maximum error of 0.0101 e; the complete ownership map and boundary diagnostics are archived in the Supporting Information. This establishes the intended overlap reconstruction route for the electronic closure and provides a molecular-scale accuracy benchmark for fragment assembly.

3.4 Quantum observables as state updates

The perceived state and the quantum calculation form a bidirectional feedback loop. An initial geometry is first perceived from SMILES or Cartesian coordinates. A single L0 calculation at that guess supplies CM5 charges and the total Mayer matrix. For an unrestricted state it additionally supplies the signed spin-selective Mayer matrix

$$B_{AB}^{\alpha-\beta} = 2 \sum_{\mu \in A} \sum_{\nu \in B} \left[(P^{\alpha} S)_{\mu\nu} (P^{\alpha} S)_{\nu\mu} - (P^{\beta} S)_{\mu\nu} (P^{\beta} S)_{\nu\mu} \right], \tag{11}$$

in the same atomic-orbital partition used for the total Mayer order. $B_{AB}^{\alpha-\beta}$ vanishes for a restricted closed shell and changes sign when the spin labels are exchanged. It measures spin-selective covalency on a bond and complements the atomwise Hirshfeld $\alpha - \beta$ population.

The CM5 vector updates electrostatic environments, the total Mayer matrix updates bond-order components and aromatic conjugation, and the $\alpha - \beta$ quantities identify where an open-shell guess carries its spin. These observables are attached to the persistent state and used to regenerate or refine the geometry with the same accepted topology. The refined structure is then perceived again; only after topology, charge, aromaticity and spin audits pass does the production electronic-structure calculation start. Thus the operational sequence is

$$\begin{aligned} &\text{guess} \longrightarrow \text{perception} \longrightarrow \text{L0 CM5/Mayer} \longrightarrow \text{geometry update} \\ &\longrightarrow \text{re-perception} \longrightarrow \text{production QM.} \end{aligned}$$

Because the total and spin-selective Mayer maps vary continuously, modest geometrical changes update this loop without forcing a premature change of chemical identity.

Hydrogen bonding is then separated from the intrinsic monomer charge while covalent identity remains fixed. The construction is formulated as a fluctuation about a CM5 population evaluated at a declared reference geometry $\mathbf{R}_{\text{ref}}$. For every accepted D–H$\cdots$A contact c, the dimer library supplies a unit response vector $\Delta\mathbf{q}_c$, including separately donor polarization, acceptor polarization and charge transfer. If the reference geometry already contains hydrogen bonds, its CM5 values include those effects. The hydrogen-bond-free baseline is recovered by

$$q_i^0 = q_i^{\text{CM5}}(\mathbf{R}_{\text{ref}}) - \sum_c s_c(\mathbf{R}_{\text{ref}})\Delta q_{i,c}. \tag{12}$$

At an arbitrary geometry,

$$q_i(\mathbf{R}) = q_i^{\text{CM5}}(\mathbf{R}_{\text{ref}}) + \sum_c \left[s_c(\mathbf{R}) - s_c(\mathbf{R}_{\text{ref}})\right] \Delta q_{i,c}. \tag{13}$$

Equation 13 reproduces the input CM5 charges at $\mathbf{R}_{\mathrm{ref}}$ and tends continuously to q_i^0 as the relevant contacts dissociate. A zero-contact reference gives $q_i^0 = q_i^{\mathrm{CM5}}$.

The reference CM5 vector already contains the electronic response of its molecular environment. Additional polarization and charge transfer are assigned to recognized events, beginning with hydrogen-bond formation. Changes in coordination, bond order, ion pairing, metal–ligand contacts, ring closure or local descriptors can identify further candidates; quantitative response is activated only when represented in the resident library.

Donor and acceptor polarization are separately fragment neutral; the charge-transfer vector is neutral only over the complete complex. The response is confined to the active X–H bond and, where present, the local C=O or N–C=O conjugated unit. Van der Waals radii determine its radial amplitude:

$$s_{\mathrm{HB}}(r) = \frac{\exp[-r/(R_{\mathrm{vdW},H} + R_{\mathrm{vdW},A})]}{\exp[-r_{\mathrm{ref}}/(R_{\mathrm{vdW},H} + R_{\mathrm{vdW},A})]}. \tag{14}$$

The correction vanishes outside the selected contact domain, where q_i smoothly returns to q_i^0. Within that domain, the resident table retains the chemically transparent 4×5 product of four donor classes (O–H, N–H, P–H and S–H) and five acceptor classes (O, N, carbonyl O, P and S). The table averages 320 rigid PBE0/def2-TZVP dimers. Four distinct donor environments and four distinct acceptor environments are crossed within each of the 20 donor–acceptor classes, giving 16 chemically audited observations per executable rule. The substituted series spans alcohols, amines, carbonyl compounds, phosphines and thiols while preserving the same fixed-geometry protocol.

The averaging preserves the physical decomposition by treating charge transfer, donor-local polarization and acceptor-local polarization separately, after which the canonical neutral atomwise vectors are rescaled to the corresponding mean displaced charges. Across the table, mean Mayer H$\cdots$A orders range from 0.031 to 0.109 and mean charge-transfer magnitudes from 0.011 to 0.078 e. The within-class sample standard deviations are 0.0018–0.014 e

for donor polarization, 0.002–0.008 e for acceptor polarization and 0.005–0.017 e for charge transfer. This spread is chemically informative: it measures the effect of substitution within a donor–acceptor class and is retained in the table alongside the uncertainty in its mean. Figure 4 shows the means and their bootstrap uncertainty, while the Supporting Information table also reports the sample dispersion and provides the individual observations, bootstrap 95% confidence intervals, leave-one-environment-out sensitivities and signed vectors. Across all response components, the largest absolute bootstrap half-width and leave-one-environment-out shift are 0.008 and 0.009 e, respectively. Thus the balanced factorial design resolves real environmental dispersion while keeping the executable class means stable on the absolute charge scale. The resulting class means are stable at the resolution supported by this factorial design and are available as executable parameters for constructing hydrogen-bond-adjusted charges from CM5 populations.

Water uses the noniterative missing-coordination and charge-transfer model introduced in Equations 18–20 of Ref. 26, replacing its integer hydrogen-bond counts by the continuous contact strength and its isolated-water reference by the symmetry-projected L0 CM5 vector $(-0.674, +0.337, +0.337)$ e. This is the water specialization of Equations 12 and 13. The published condensed endpoint is the ForceBalance-refined three-point transferable intermolecular potential (TIP3P-FB) vector $(-0.848, +0.424, +0.424)$ e, and the published water–water transfer scale is $\beta = 0.10$ e. The two donor and two acceptor capacities share the CM5-to-TIP3P-FB polarization increment symmetrically. Consequently, isolated water is exactly CM5, whereas the central molecule of the symmetric pentamer, which donates and accepts two hydrogen bonds, is exactly TIP3P-FB and exactly neutral. The two donated and two accepted charge-transfer contributions cancel at that endpoint.

The 4×5 table is a versioned runtime resource of the common chemistry layer and a directly executable product of the validation. A single shared application programming interface (API) accepts connectivity, geometry and either intrinsic or environment-polarized CM5 charges, then returns tuned charges together with separate polarization, charge-transfer

and contact diagnostics. Repeated optimization, exploration and simulation steps reuse the same recognition plan and persistent skinned pair list. At a nonperiodic boundary, the same closed-form rule supplies the missing neutral polarization. The published water specialization and the general dimer-derived CM5 extension remain distinct, auditable and available throughout the workflow. The corresponding electrostatic construction is illustrated in Figure 5.

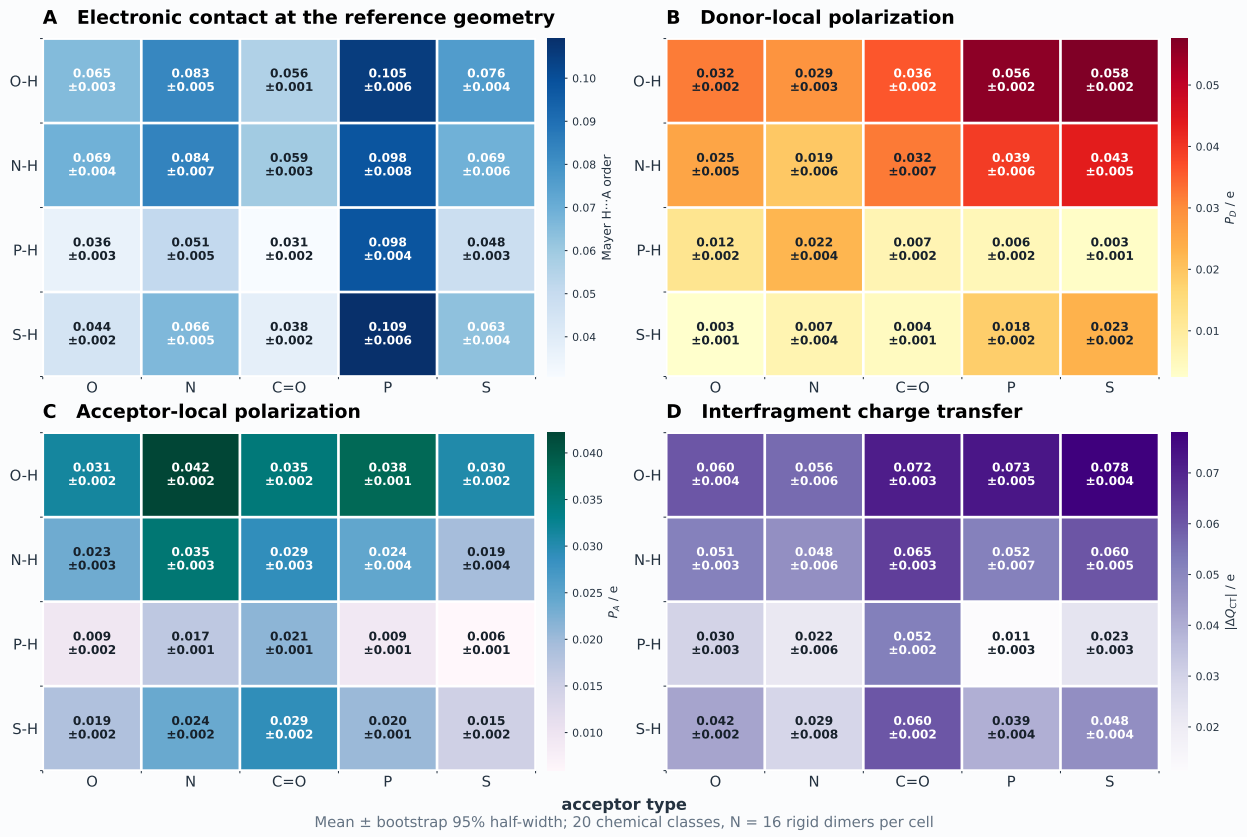


Figure 4: Mean hydrogen-bond responses across 20 donor–acceptor classes.

3.5 Geometry and population decision path

For conventional organic systems, the practical objective is a reproducible path from a SMILES string to an accurate geometry. Transition-metal systems start from the enriched Cartesian state defined above or from a coordination specification that first generates candidate Cartesian structures. Only three level labels are needed for this optional route. Beyond

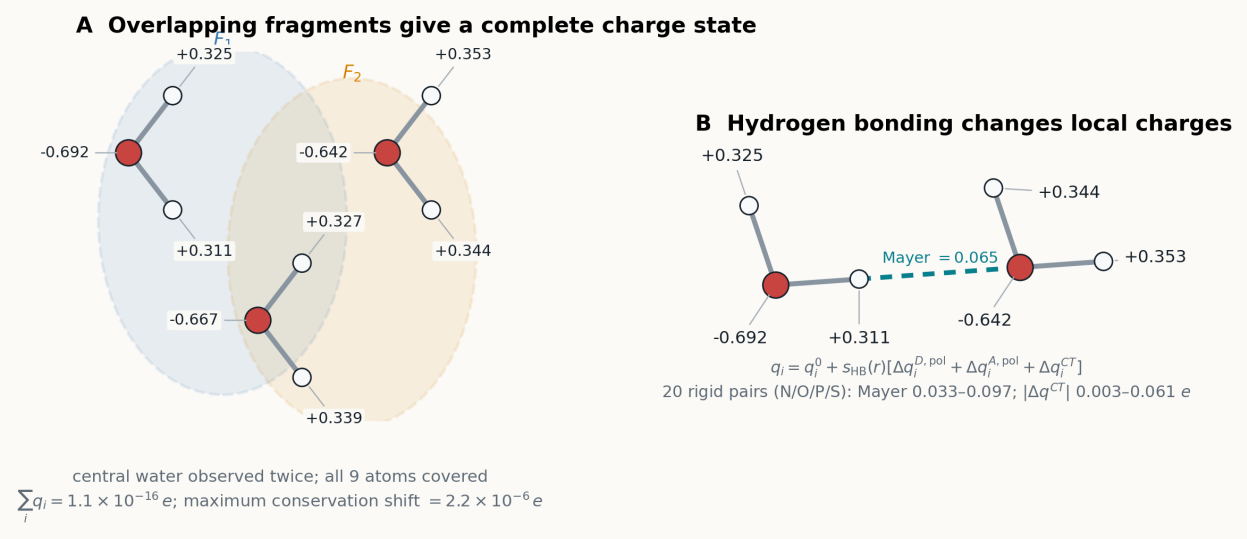


Figure 5: Baseline electrostatic closure from charge assembly and local hydrogen-bond response.

these three active labels, two additional archival or reserved identifiers appear in file names and figures. The baseline introduced here (L0) is PBE0/def2-TZVP; a higher-level geometry (L1) is obtained with a double-hybrid method and an extended basis set, and PL1 applies a further posterior correction to covalent bond lengths and hydrogen-bond geometries. These levels retain the definitions established in the structural-accuracy ladder and its underlying high-level, templating and posterior-correction developments.[15,27–30] When the full route is requested, the workflow proceeds incrementally,

$$\text{molecular input} \longrightarrow \text{initial Cartesian structure} \longrightarrow \text{L0} \longrightarrow \text{L1} \longrightarrow \text{PL1}, \tag{15}$$

reusing the approximate curvature accumulated by each optimization to initialize the next. If a Cartesian file is supplied, its geometry is preserved unless optimization is explicitly requested, in which case the same ladder starts from that structure.

Population analysis and geometry optimization remain independent requests. Accordingly, the water, hydrogen-bond, ferrocene and cisplatin calculations reported here are fixed-geometry single points. When a larger target is requested, compatible fragments from the published 2025 building-block library and its 2026 extension are reused whenever possible;

otherwise, connected overlapping fragments are generated until every atom and covalent pair has been defined. The Supporting Information gives the corresponding input details, the reuse of Hessian information between stages and the complete decision tree.

L0 is an electronic closure and feedback route that complements geometry-only perception. Geometry-only calls accept a complete state or request topology, rings, symmetry and continuous geometric descriptors. L0 supplies the reproducible CM5/Mayer/spin observables used by the feedback examples. Functional and basis sensitivity, together with the geometry-only baseline, define the next expansion of the electronic benchmark.

The current ORACLE implementation also supports an explicitly requested single-geometry transition-state perception. Weak acyclic contacts that may represent breaking interactions and near-covalent contacts that may represent forming interactions are retained as typed transition-state evidence, but are not silently promoted to the constitutional graph. The TS contract records the reactive kernel, the allowed chart policy (reactive distance or prescribed pseudobond), the reactive zones, and the provenance of every decision. When several admissible primitive combinations describe the same reactive zone, the candidate charts are compared through the normalized condition number of their Wilson-B representation. Provided the declared conditioning trigger and minimum relative-gain gate are satisfied, ORACLE selects the candidate with the lowest condition number and records the comparison and selection. Thus a formally complete chart is not accepted merely because it has full rank: among valid alternatives, numerical conditioning is an explicit chemical-workflow criterion.

The examples below illustrate this feature through representative transition-state geometries with insertion, ring-opening, dissociation, torsional-isomerization and rearrangement topologies, including two rank-24 bicyclobutane structures (Figure 6).

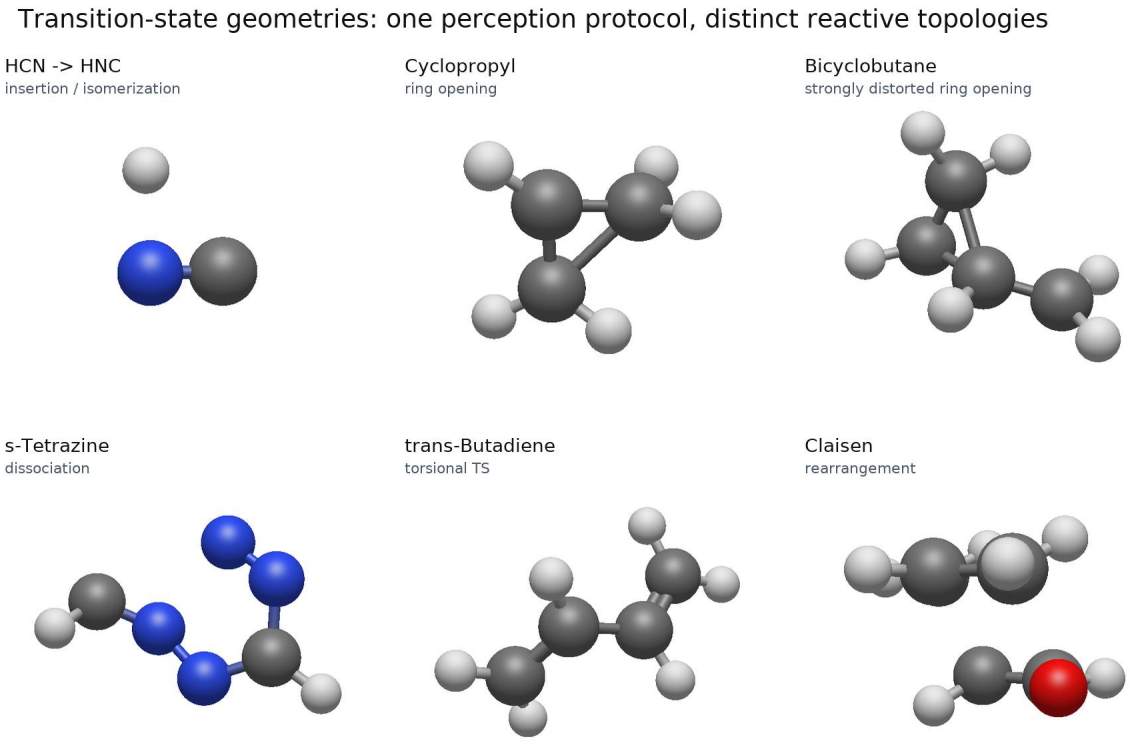


Figure 6: Transition-state geometries spanning distinct reactive topologies.

3.6 Executable symmetry

Symmetry becomes executable by storing each accepted operation as

$$g = (\mathbf{R}_g, \pi_g), \qquad \mathbf{R}_g^{\mathsf{T}}\mathbf{R}_g = \mathbf{1}, \qquad \det \mathbf{R}_g = \pm 1, \tag{16}$$

where π_g is the element-preserving atom permutation induced by $\mathbf{R}_g$. Identity, closure, inverse, matrix–permutation composition and mapping residuals are verified jointly. The complete action can therefore be applied directly to atoms, polar vectors, axial vectors or primitive coordinates.

Degenerate inertia tensors receive a deterministic, rigidly covariant treatment. For a symmetric top, the unique principal axis is fixed by the inertia tensor and the free perpendicular plane is anchored to an actual point-set operation: either a perpendicular twofold axis or the normal to a vertical mirror plane. Spherical tops instead recover the complete point-set isometry from element- and metric-compatible triples. Accepted matrices are conjugated back to the input Cartesian frame before serialization, so neither the group nor its atom action depends on an arbitrary eigensolver basis.

Finite groups are required to have their exact order and to close jointly in matrix and permutation space. This gate covers the low-order groups C_1, C_s and C_i; the complete supported axial families C_n, C_{nv}, C_{nh}, D_n, D_{nh} and D_{nd}; the pure improper families S_{2n}; and the polyhedral families T, T_d, T_h, O, O_h, I and I_h. Linear molecules are treated separately as $C_{\infty v}$ or $D_{\infty h}$ and are not subjected to an artificial finite-group closure.

The serialized operation–permutation pairs are also the sole molecular symmetry authority passed to coordinate construction. ORACLE uses the same operation classification, and character construction uses the stored matrices whenever a label alone is insufficient. The workflow can therefore build every irreducible representation and its symmetry-adapted linear combinations (SALCs) from the complete ORACLE action, within homogeneous coordinate families and without selecting a convenient subgroup or reconstructing symmetry

independently.

Quasi-symmetry uses an absolute Cartesian threshold τ, a stable-threshold plateau and an explicit residual margin. If a complete group G is accepted, the geometry may be projected by

$$\mathbf{r}_i^{\mathrm{sym}} = \frac{1}{|G|} \sum_{g \in G} \mathbf{R}_g^{\mathsf{T}} \mathbf{r}_{\pi_g(i)}. \tag{17}$$

The threshold, operation residuals and required Cartesian displacement remain part of the state. Symmetry projection follows acceptance of the complete group under the declared threshold.

Projection is a coordinate transformation with a recorded displacement. The quasi-O_h cubane audit gives a maximum displacement of 3.3×10^{-4} Å and reanalysis restores the 48 operations to a residual below 10^{-15} Å. Frequencies, Derivative properties are evaluated on the projected geometry whenever the recorded displacement is chemically relevant.

3.6.1 Assignment and adjustment of almost symmetric structures

Assignment tests whether the observed point set supports a complete finite group within a declared absolute tolerance. Adjustment applies the group-average projector only after every operation has a compatible atom permutation and the set has the expected order. An incomplete group would destroy the projector property; the diagnostic is recorded and the structure is left unchanged.

This separation is essential for experimental, edited, force-field and unconstrained optimized geometries, which often retain symmetry only approximately. Water and formaldehyde regressions identify stable tolerance plateaus, store the complete C_{2v} permutations, project, and reanalyze the adjusted structures. The output records both the assigned symmetry and the Cartesian change required to make it exact.

Exact molecular symmetry is complemented by local regularity because medium-to-large molecules are often globally C_1 while a coordination sphere, substituted ring or repeated fragment remains approximately symmetric. Continuous elemental, radial and local-environment

descriptors order these domains and define reproducible local equivalence classes. The stored molecular group provides global equivalence.

For applications that change geometry, the workflow internally generates ordered primitive coordinates and their analytic Wilson matrix from the frozen state. They support reconstruction and correction internally; their definitions and rank tests are given in the Supporting Information.

3.7 Persistent aromaticity and differential structural correction

Aromaticity is stored as more than an atom flag. For each accepted ring, the state records the contributing atoms and bonds, plane deviations, electron count when available, and the decision criterion: $4n + 2$, a fused conjugated component, or a metal-coordinated η^5 cycle. Local p-orbital eligibility includes common group-13, -14, -15, and -16 centers. During structural correction, every stored aromatic ring supplies a local best-fit-plane projector, with proposals for shared atoms averaged consistently. A helically arranged fused system therefore retains distinct local ring planes throughout projection.

Given a target geometry calculated at level L and a matched accurate fragment f, the correction transfers each supported primitive as

$$q_\alpha^{\mathrm{corr}}(M; L) = q_\alpha(M; L) + [q_\alpha(f; \mathrm{ref}) - q_\alpha(f; L)]\,. \tag{18}$$

The molecule and every selected fragment use the same level L, so the bracket isolates the method-specific error. Equation 18 applies simultaneously to stretches, bends, torsions and out-of-plane coordinates. Environments outside the resident library become explicit requests for targeted fragment calculations. Back-transformation is followed by the persistent aromatic and finite-group projectors and complete reperception.

A local target displacement $\Delta\mathbf{q}_S$ is reconstructed by the regularized update

$$\Delta\mathbf{x} = \mathbf{B}_S^{\mathsf{T}} \left(\mathbf{B}_S\mathbf{B}_S^{\mathsf{T}} + \lambda\mathbf{I}\right)^{-1} \Delta\mathbf{q}_S. \tag{19}$$

The examples below use this operation for core–valence, conjugation and selected-pair corrections. Structural correction consumes the perceived state through a downstream model that supplies parameter construction, validation, coordinate selection and force-field fitting.

4 A Validation Designed Around State Transfer

The validation mirrors the structure of the central claim. It first tests invariance to rigid motion and atom ordering, then verifies exact recovery of topology, cycle rank, aromaticity, finite groups, primitive rank and serialization, and finally examines reuse in chemically distinct applications with explicit reference coverage. Inputs, fingerprints, scripts and machine-readable outputs are documented in the Supporting Information. The reported tests quantify reproducibility and internal consistency, while frozen inputs and executable records make each contract independently testable.

As a secondary efficiency check, we performed a controlled handoff ablation on eight systems spanning 3–176 atoms. The full route supplied a validated ORACLE state, whereas the ablated route supplied the identical Cartesian source and forced the downstream frontend to reconstruct that state. All input structures and downstream settings were held fixed. The persistent route was faster in all eight cases (median 0.257 versus 0.327 s; median case-wise speed-up 1.33-fold), while the accepted chemical representation and downstream results were identical in every case. This quantifies the computational benefit of state reuse at the workflow handoff. The held-out thiazole workflow measures structural transfer through successive perception, retrieval and geometric-correction stages, where preserving the accepted state determines which chemical objects and primitives are transferred. This test isolates the Efficiency claim in Table 5: computational reuse and handoff consistency under an un-

changed chemical result. It is distinct from the Capability tests (transition-state geometries and $Fe_4S_4Cl_4$), which exercise tasks unavailable to a graph-only route, and from the Differentiation test (state-blind control), which isolates the effect of retaining electronic state at fixed geometry.

The Python implementation provides the portable reference, while the dominant graph, continuous-perception and cycle-space kernels also have compiled implementations that release the Python global interpreter lock. Both paths must return the same accepted graph and canonical cycles. Build information, fallback rules and cross-backend tests are reported in the Supporting Information.

The same persistent state supplies topology, symmetry, primitive coordinates and electronic descriptors to downstream optimization and returns with the updated geometry for dependency-aware reanalysis.

5 Reproducible Computational Realization

The implementation connects molecular input, continuous perception, quantum- mechanical observables and downstream coordinate construction through one versioned state. Chemoinformatics and electronic-structure components are interchangeable execution layers; the chemical state remains the source of identity, provenance and transfer. The complete computational specification, software versions and input fingerprints are provided in the Supporting Information.

6 State Transfer Across Chemical Regimes

The results are organized as state-transfer experiments. They follow the accepted identity through irrelevant perturbations, executable discrete objects, application boundaries and explicit coverage diagnostics. The most demanding tests are the chemically discriminating examples: transition states, open-shell systems, interactions, coordination complexes and a

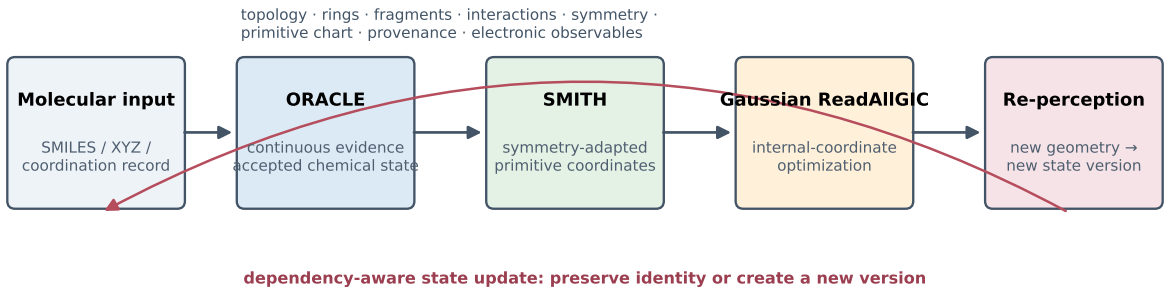


Figure 7: State transfer from perception through coordinate construction and re-perception.

Table 4: Four discriminating tests of the persistent chemical state.

Test	What it establishes
Transition-state geometries (25 cases)	One local perception/chart protocol spans ring opening, dissociation, insertion, torsional isomerization and rearrangement.
$Fe_4S_4Cl_4$ (C_{3v})	Explicit coordination state preserves a non-obvious 22-edge polynuclear topology and the rank of the nonredundant internal coordinates.
Seven-target leave-one-out transfer	SWITCH, ORACLE and the primitive chart transport reference information to held-out structures; thiazole gives the representative quantitative case.
State-blind electronic control	At fixed nuclear geometry, graph-only output is unchanged across state and environment changes, whereas ORACLE updates CM5, Mayer and spin observables.

held-out structural correction. Full numerical tables and fitted-model diagnostics are consolidated in the Supporting Information. The state-transfer sequence is depicted in Figure 7. The four discriminating tests are summarized in Table 4.

Two chemically demanding demonstrations anchor the validation. The transition-state examples test whether one perception protocol can span ring opening, dissociation, insertion, torsional isomerization and rearrangement. $Fe_4S_4Cl_4$ tests whether the same state can encode a polynuclear coordination topology whose 22 accepted edges are not recoverable from simple organic-valence rules. Figures 6 and 10 place these two cases at the center of the subsequent quantitative audits.

6.1 One state survives geometric and representational stress

Figure 8 summarizes the elementary validation through eleven predeclared topology cases spanning isolated, fused, spiro, cage and metallocene graphs. In every case ORACLE recovers the expected bond count, cycle-space rank, elementary cycle sizes, aromatic-atom count and

Table 5: Evidence hierarchy used in the validation.

Evidence class	Comparison	Interpretation
Efficiency check	Persistence versus forced re-perception	Same chemical result with modestly faster state reuse.
Capability	Transition-state geometries and $Fe_4S_4Cl_4$	Tasks and topologies represented natively in the ORACLE state.
Differentiation	State-blind fixed-geometry control	Same graph, state-dependent CM5/Mayer/spin output.

connected-fragment count. The aromaticity rule combines planarity, local p-orbital eligibility and $4n + 2$ counting, with separate criteria for fused conjugated components and metal-coordinated η^5 rings. Aromatic assignment requires the complete set of chemical criteria.

Seven molecular benchmarks from C_{2v} to I_h recover complete operation sets and atom permutations, including D_{5h}, D_{5d}, T_d, O_h and the 120 operations of I_h. A separate generic audit spans 45 finite groups: all six axial families for $n = 2$–6, S_4 through S_{12}, the three low-order groups, and all seven tetrahedral, octahedral and icosahedral families. For every case ORACLE verifies identity, exact order, inverse and matrix–permutation closure after a fixed three-dimensional rotation and translation; the coordinate layer then constructs the complete finite character sequence from that same action. The intermediate D_5 ferrocene geometry is an additional discriminating regression: it recovers all ten operations, and the resulting 57 internal coordinates close the full vibrational rank.

For distorted formaldehyde, a threshold sweep distinguishes the onset of a stable quasi-C_{2v} plateau from the smaller-τ C_s result. The selected $\tau = 5 \times 10^{-4}$ Å is the smallest plateau value with adequate residual margin; projection requires only 8.07×10^{-5} Å maximum displacement and reanalysis returns exact C_{2v} to numerical precision.

The continuous descriptors are invariant to rigid motion and atom order at numerical precision for water, benzene and methane. A sampled 2×10^{-4} Å local displacement produces a correspondingly small descriptor response while the discrete topology remains unchanged, completing the local regression contract around the sampled geometries.

An additional ethanol check connects low- and high-information states. Complete-link clustering of the seven descriptor components yields five transparent classes—the two carbon

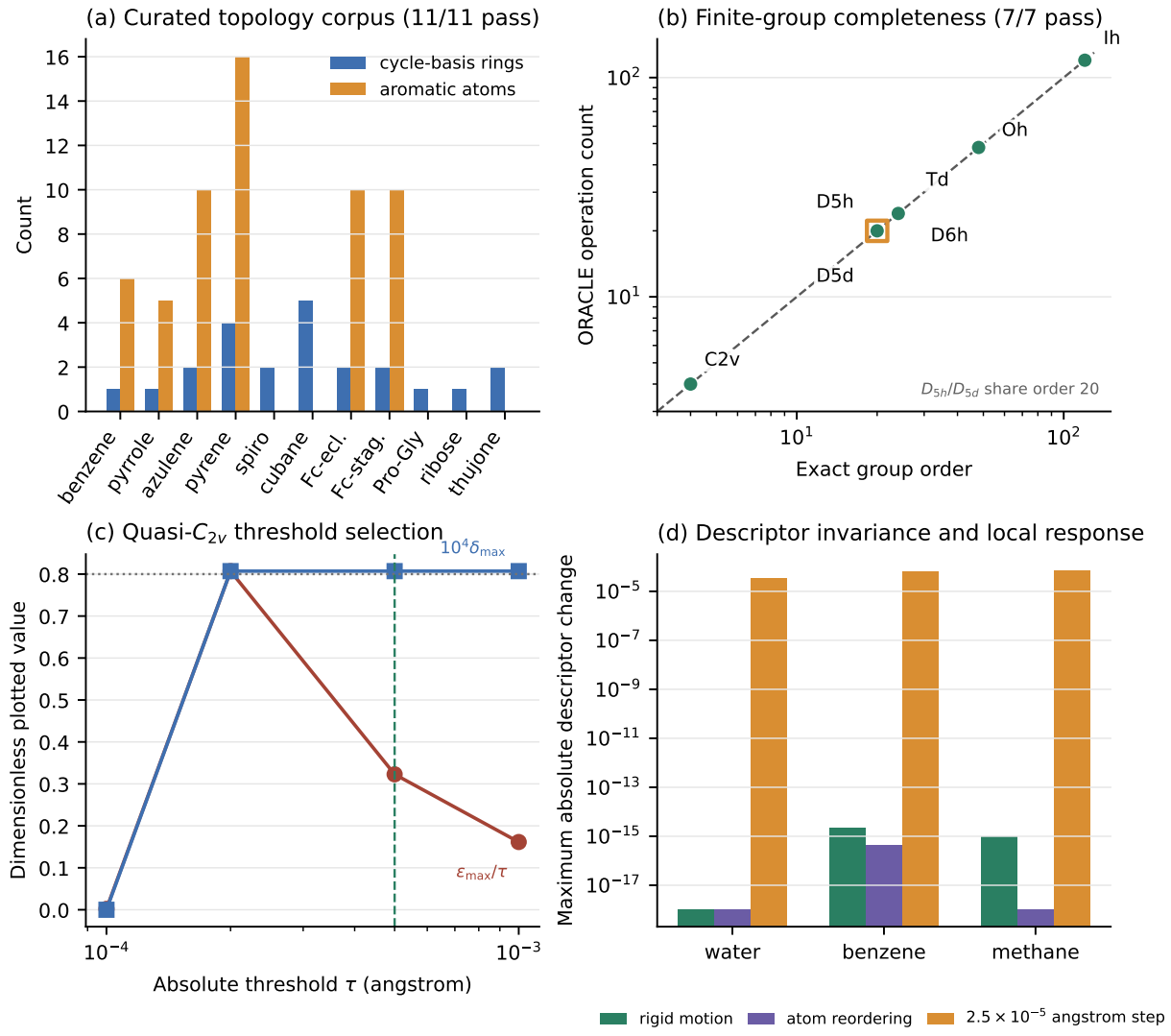


Figure 8: Validation of topology, aromaticity, finite-group action, quasi-symmetry and descriptor invariance.

environments, oxygen, carbon-bound hydrogen and hydroxyl hydrogen—and those classes remain unchanged from one quarter to twice the declared thresholds as well as after atom-order reversal. The result illustrates an availability-level contract that remains auditable across different information levels.

6.2 The state is executable at practical scale

The compiled path keeps the additional chemical evidence affordable. Pair geometry, continuous refinement, accepted-graph construction, cyclic-core restriction and independent-cycle selection are evaluated together, while lazy materialization avoids dense matrices when a consumer requires topology alone. Relative to the equivalent Python reference, this path is 5.74-fold faster at the median over six systems, ranging from 2.24-fold for water to 9.93-fold for paclitaxel; on the reported arm64 host, the vancomycin time decreases from 36.5 to 4.0 ms. The compiled and Python paths return identical edges and canonical cycles, the wider audit retains cyclic-core/ Horton equivalence for 4991/4991 molecules, and all graph, ring-rank and benchmark checks pass for 346/346 structures. Exact environments, raw timings, compiled-build metadata and the Python fallback are documented in the Supporting Information.

An independent RDKit comparison places the graph and cycle machinery on the scale of current cheminformatics practice.[1] Agreement counts and disagreement records are reported in the Supporting Information. The comparison establishes graph and cycle conformance for equilibrium structures. The transition-state examples extend this representation by retaining forming and breaking contacts as typed evidence and by selecting among primitive charts through the Wilson-B condition number before downstream coordinate construction.

6.3 Quantum mechanics feeds the chemical state back

Three complementary chemical domains make the perception–QM connection concrete. Aromatic systems test how total Mayer orders refine conjugation, open-shell systems add the

spatial distribution of spin, and transition-metal complexes test the same population contract across all-electron and effective-core-potential descriptions.

Chemical discrimination is examined separately on a stratified corpus of 58 frozen structures spanning benzenoid, heteroaromatic, fused, nonbenzenoid and boundary systems together with nonaromatic controls. Every geometry passes the declared numerical gate. The present method and geometry-first RDKit return identical aromatic-atom sets in 58/58 cases; the same agreement is obtained when RDKit receives the accepted covalent graph. The exact-match fraction is therefore 1.00, with a two-sided Wilson 95% interval of 0.94–1.00, and the mean and minimum Jaccard indices are both 1.00. Cyclopropenone and acenaphthylene are included explicitly as discriminating boundary cases. The result measures conformance between independent implementations over the declared corpus. The full atom sets, strata, source hashes and environment are reported in the Supporting Information.

The electronic-feedback route is tested independently on 18 neutral doublets distributed over nine electronic classes: localized, unsaturated and conjugated cyclic carbon radicals, heteroatom-substituted carbon radicals, aryloxy, carbonyl, nitroxide, nitrogen-centered and peroxy systems. All unrestricted Kohn–Sham PBE0 (UKS-PBE0)/def2-TZVP L0 calculations converge; the Hirshfeld $\alpha - \beta$ populations satisfy the unit-spin sum exactly after the declared 2×10^{-4} quadrature gate, and $\langle S^2 \rangle$ ranges from 0.754 to 0.805. The absolute-spin participation spans 1.38–4.85 atoms, with a bootstrap 95% interval of 2.00–2.96 atoms for its corpus mean. Maximum absolute $\alpha - \beta$ Mayer components span 0.002–0.46, with a corresponding bootstrap mean interval of 0.068–0.23. In phenyl, the six ring bonds have total Mayer orders of 1.408–1.514 while their spin-selective components remain at or below 0.047; the L0 state therefore separates aromatic bond equalization from the localized open-shell response. Together, the nine-class stratification and bootstrap intervals provide a quantitative characterization of the declared open-shell set, with paired observations resolving the electronic response within each class. They establish the reverse, quantum-mechanical leg of the perception–QM connection: electronic observables update the chemical state before

geometry refinement and the production calculation. Functional conformance is additionally checked on macOS/arm64 and Linux/x86-64, with environment-specific records; timings are compared only within the same host.

State-specific electronic observables provide a direct test of the bidirectional contract. On the same ORACLE-perceived geometry, CH_2 was evaluated as both a singlet and a triplet, and O_2 as a triplet. The CH_2 C–H Mayer order changes from 0.938 to 0.951 and the carbon CM5 charge from -0.179 to -0.201 e on changing the multiplicity. A complementary time-dependent density-functional theory in the Tamm–Dancoff approximation with PBE0 (TDA/PBE0) gives a singlet S_1 excitation at 3.98 eV; the carbon and oxygen CM5 charges change from $+0.053$ and -0.290 e in the ground state to -0.307 and $+0.003$ e, while the C=O Mayer order decreases from 2.098 to 1.445. These state-dependent changes illustrate why CM5/Mayer observables can influence subsequent atom typing, charge assignment and force-field parameter selection. For uracil, the two carbonyl Mayer orders change from 2.072 and 2.037 in S_0 to 1.966 and 1.828 in S_1; the S_0 values become 2.049 and 1.977 in aqueous conductor-like polarizable continuum model (C-PCM). The C4=O4 carbonyl is more strongly perturbed in both comparisons, demonstrating state- and environment-dependent parameter selection. The complete records are provided in the Supporting Information.

The fixed-geometry comparisons define a direct boundary with graph-only cheminformatics. Changing multiplicity, electronic state or dielectric environment does not change the nuclear graph, so a conventional graph perception layer would return the same structural object. ORACLE retains the electronic and environmental context and propagates the resulting CM5, Mayer and spin observables into the chemical state. This is the mechanism by which non-obvious open-shell states, excited states and solvent-aware parameter selection remain distinguishable downstream.

A three-level sensitivity calculation quantifies the range of the L0 closure. At the same ORACLE geometries, PBE0/def2-SVP, PBE0/def2-TZVP, and B3LYP/def2-TZVP give formaldehyde S_1 C=O Mayer orders of 1.459, 1.445, and 1.430, respectively, while the

corresponding S_0 values are 2.106, 2.098, and 2.092. For uracil, the two S_1 carbonyl orders are 1.966/1.828 at PBE0/def2-TZVP, compared with 2.072/2.037 in S_0; the functional/basis range is reported in the Supporting Information. The state effect is therefore larger than the level variation in these examples, while the level variation remains large enough to justify preserving the declared functional, basis, and solvent metadata with the chemical state.

The state-blind control holds the nuclear geometry and covalent graph fixed and assigns no state-dependent electronic observables. Its structural output is therefore unchanged, whereas ORACLE records a carbon CM5 change of $-0.022\ e$ and a mean C–H Mayer change of $+0.013$ for CH_2 singlet to triplet. For formaldehyde S_0 to S_1, the corresponding carbon and oxygen CM5 changes are -0.360 and $+0.293\ e$, and the C=O Mayer change is -0.652. The control isolates the scientific consequence of retaining electronic state at fixed geometry and shows how the persistent state records changes that a geometry-only route leaves unrepresented. The numerical consequences are collected in Figure 9 and Table 7. This is not a performance comparison between competing implementations of the same electronic task: standard graph-only toolkits do not compute Mayer spin components or state-specific electronic observables. The comparison therefore tests whether the chemical state retains information that a graph-only object cannot contain.

At fixed nuclear geometry, any perception layer limited to the covalent graph—including RDKit, Open Babel, or the Chemistry Development Kit (CDK) in their standard operation—returns an unchanged structural object for all four comparisons in Table 6. ORACLE differentiates them because electronic-state and dielectric metadata are part of the persistent object; the resulting CM5, Mayer, and spin changes are therefore state evidence rather than a geometry-only modeling choice.

The fixed-geometry electronic comparison is illustrated in Figure 9 and quantified in Table 7.

The transition-metal examples complete this electronic picture (Figure 11). All-electron D_{5h} ferrocene gives $q_{\mathrm{Fe}} = +0.984\ e$ and ten equivalent Fe–C Mayer orders of 0.441. With

Table 6: State-dependent chemical evidence at fixed nuclear geometry.

Case	CM5 change	Mayer change	State object	Downstream use
CH_2 $S_0 \rightarrow T_1$	$\Delta q_C = -0.022\ e$	$\Delta BO_{C-H} = +0.013$	synthon/Z_i^{eff} update	spin-aware atom typing and parameter routing
H_2CO $S_0 \rightarrow S_1$	$\Delta q_C = -0.360\ e$	$\Delta BO_{C=O} = -0.653$	carbonyl synthon update	excited-state bond and force-field selection
Uracil $S_0 \rightarrow S_1$	state-specific vector	$\Delta BO_{C4=O4} = -0.209$	carbonyl synthon/Z_i^{eff} update	inequivalent carbonyl parameter selection
Uracil gas $\rightarrow$ C-PCM	solvent-specific vector	$\Delta BO_{C4=O4} = -0.060$	environment-aware synthon	solvent-aware equilibrium parameters

Table 7: State and electronic-level sensitivity at fixed nuclear geometries.

Case	PBE0/def2-SVP	PBE0/def2-TZVP	B3LYP/def2-TZVP
CH_2 S_0: C–H Mayer	0.944	0.938	0.936
CH_2 T_1: C–H Mayer	0.969	0.951	0.951
H_2CO S_0: C=O Mayer	2.106	2.098	2.092
H_2CO S_1: C=O Mayer	1.459	1.445	1.430
Uracil S_0: two C=O Mayer	2.135/2.067	2.072/2.037	2.081/2.040
Uracil S_1: two C=O Mayer	2.042/1.881	1.966/1.828	1.976/1.825

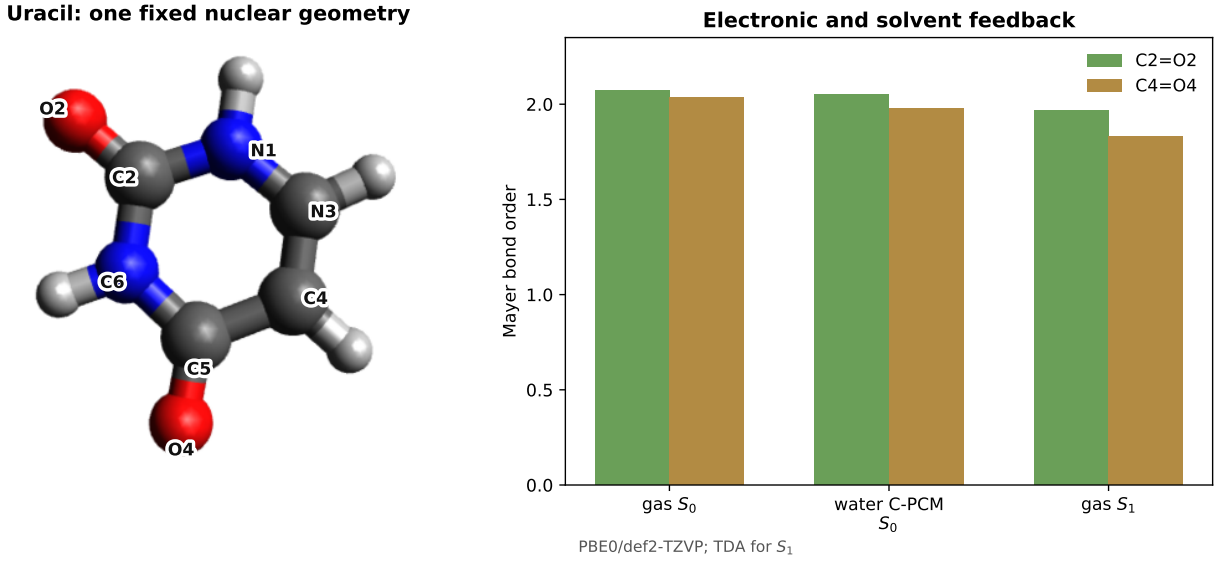


Figure 9: Uracil at fixed geometry; the C4=O4 bond is most strongly perturbed.

the native def2 effective core potential, C_{2v} cisplatin gives $q_{\mathrm{Pt}} = +0.657\ e$, Pt–Cl 0.802 and Pt–N 0.520. Both fixed-geometry calculations converge and conserve charge within $3 \times 10^{-16}\ e$, showing that the same perceived state and population analysis remain coherent across distinct coordination and relativistic regimes.

A more demanding coordination example is the neutral singlet $Fe_4S_4Cl_4$ cluster in C_{3v} symmetry (12 atoms). Its topology is not recoverable from a simple ligand-counting rule: the accepted graph contains the six Fe–Fe edges of the Fe_4 tetrahedron, twelve Fe–S edges forming the cubane-like core, and four terminal Fe–Cl edges, for 22 accepted edges in total, with no S–S or Cl–Cl edges. The enriched XYZ record stores this graph, the continuous bond-order components, and the requested C_{3v} symmetrization before coordinate construction. The independent coordinate output starts from 182 primitive candidates, filters these to 136 admissible candidates, and retains 30 frozen internal coordinates that reach the complete non-linear rank $3N - 6 = 30$; matrix–permutation closure is confirmed, with 21 totally symmetric A_1 coordinates. This case exercises the enriched coordination representation and shows why explicit connectivity, symmetry, and primitive-coordinate provenance are needed when a polynuclear topology is chemically non-obvious.

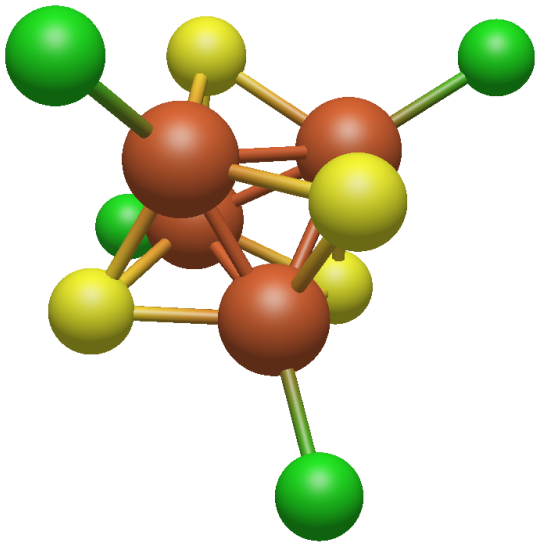


Figure 10: Neutral singlet $Fe_4S_4Cl_4$ coordination topology in C_{3v} symmetry.

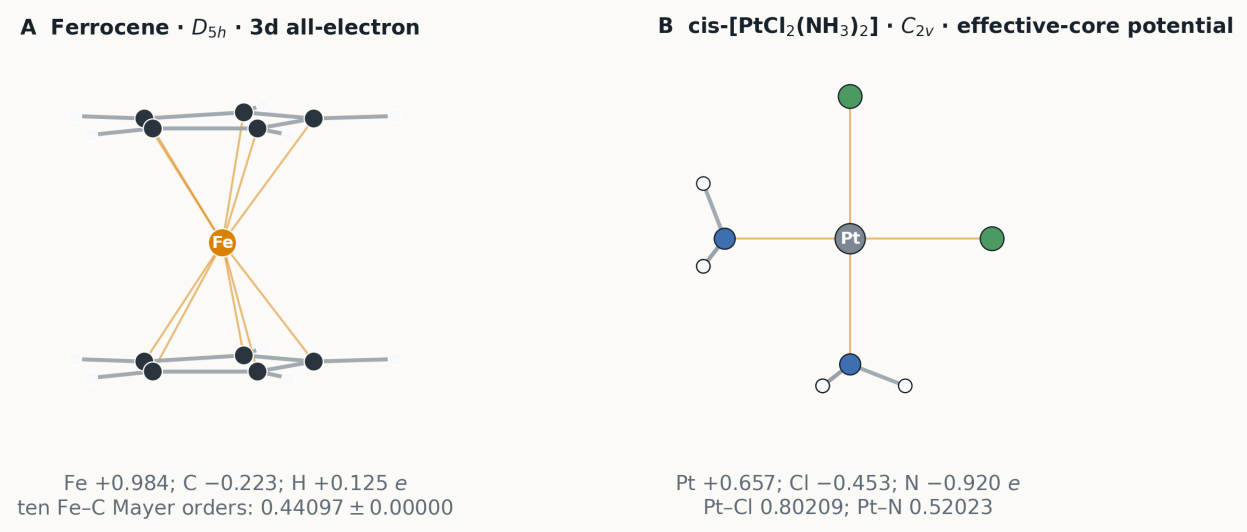


Figure 11: Baseline populations for ferrocene and cisplatin at fixed geometry.

6.4 Chemical consequences of continuous perception

The atomic synthon is the local chemical object that carries continuous evidence into subsequent operations. Its effective atomic number provides a compact coordinate for comparing environments while preserving elemental identity. This combination makes related atoms clusterable without erasing gradual changes in geometry or electronic population, and supports atom typing, fragment construction and parameter retrieval. The same descriptors remain meaningful when the nuclear geometry is unchanged but the electronic state or environment changes, as shown for CH_2, formaldehyde, uracil and gas/solvent comparisons. Reactive contacts become typed evidence for transition-state charts, while continuous coordination retains non-obvious metal environments such as $Fe_4S_4Cl_4$. The relationship between elemental identity, local environment and Z_i^{eff} is summarized in Figure 12.

6.5 Test case I: traceable [6]helicene assembly from overlapping tiles

Fragment-based construction uses shared geometry to determine relative placement. This information is essential for fused, polycyclic and stereochemically congested targets. OR-

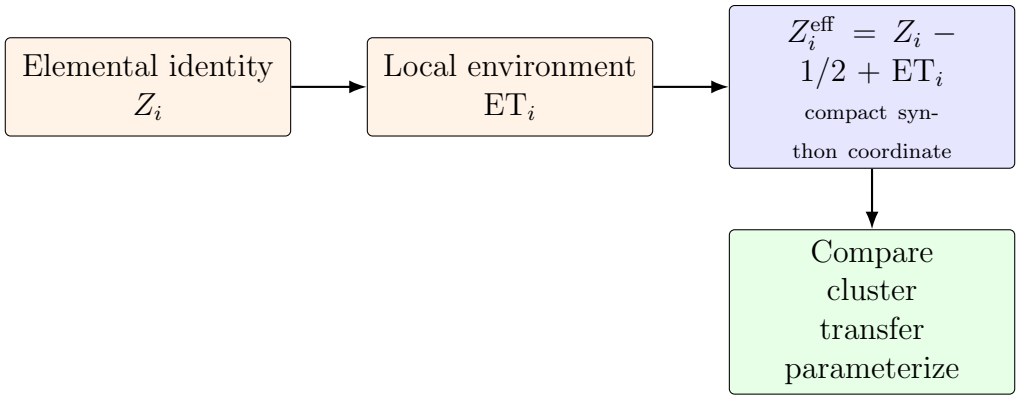


Figure 12: Role of the effective atomic number in the ORACLE chemical module.

ACLE therefore treats overlap as the geometrical contract. Every fragment after the seed must share at least three non-collinear heavy atoms with the assembled structure and add at least one new atom. The shared coordinates define a proper Kabsch rigid fit; collinear overlaps are rejected, and the source path, atom mapping, new/overlap sets and fit root-mean-square deviation (root-mean-square deviation, RMSD) are serialized. A deterministic beam search selects the covering set and a joint rigid-pose fit then optimizes all selected placements simultaneously.

[6]Helicene is a discriminating test with simple connectivity and a nonplanar three-dimensional structure. A planar polycyclic drawing contains a severe terminal steric conflict, whereas the physical structure is helical and chiral. The local cache of the published 2025 chemical building-block library (LCB25) contains a naphthalene geometry from its companion input level,[31] with 18 atoms including hydrogens, below the literal 20-atom limit. The internal SWITCH parser described in Section 5 identifies five naphthalene occurrences in the target [6]helicene graph. The first covers 10 carbon atoms; each subsequent occurrence shares a complete six-carbon ring and adds four carbons. The five placements therefore cover all 26 target heavy atoms, while every step is more strongly determined than a single-bond link (Figure 13).

The four non-seed alignments have RMSDs 0.014, 0.013, 0.013 and 0.011 Å; the joint pose fit converges in 18 iterations and passes the strict 0.35 Å quality gate cleanly. A small deter-

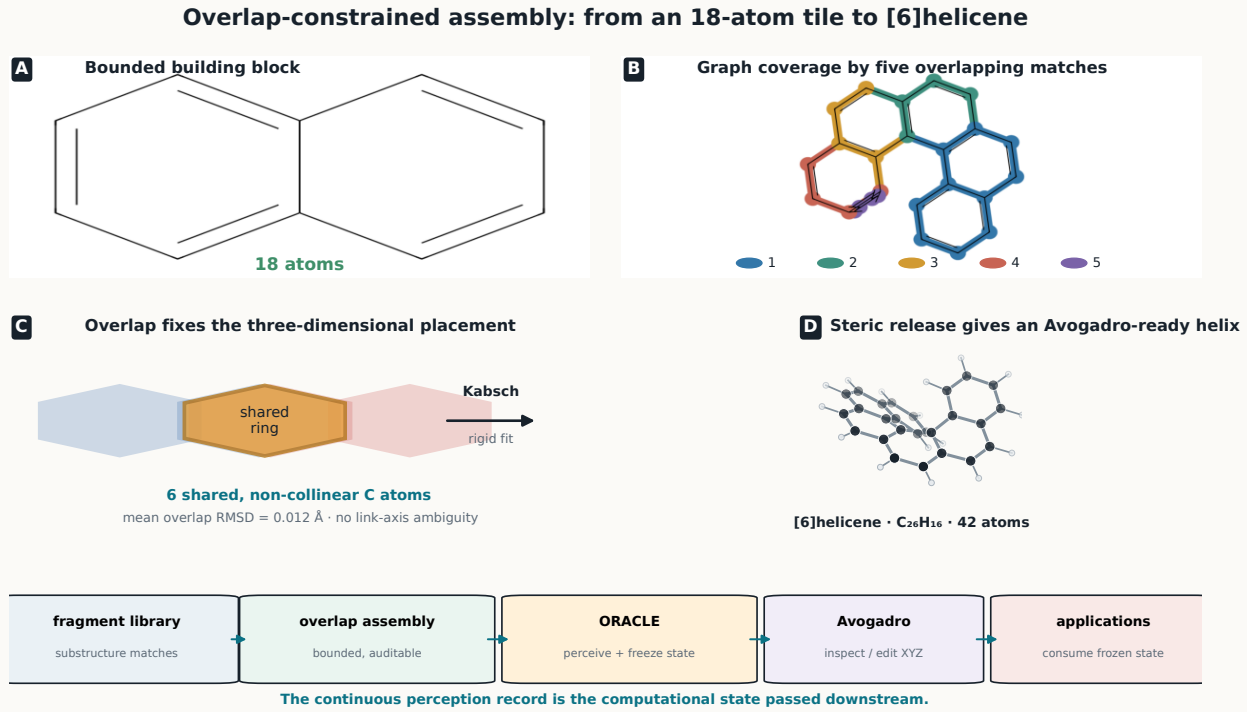


Figure 13: Overlap-constrained [6]helicene construction from five matched building-block tiles.

ministic displacement normal to the initially assembled aromatic scaffold breaks the exactly planar steric saddle; Geometry, Frequency, Noncovalent force field (GFN-FF) relaxation releases the terminal contact while preserving the fragment-derived geometry. Hydrogens are added only after complete heavy-atom coverage. The resulting $C_{26}H_{16}$ structure contains 42 atoms, and an ORACLE reimport recovers 47 covalent bonds, six rings and C_2 symmetry. The default output is strict XYZ for direct ORACLE and Avogadro use; the adjacent `oracle.xyz.assembly.v1` JavaScript Object Notation manifest stores mappings, provenance, beam-search and pose diagnostics, and the quality decision. The assembled structure is shown in Figure 13.

A reusable carbohydrate subset. The same construction also applies beyond fused aromatics. The 2026 chemical building-block library (LCB26) adds high-level equilibrium structures of representative sugars and their oxygenated fragments. The present carbohydrate-focused set contains xylose, erythrose, erythrulose, threose and fructose together with glycidaldehyde, glycidol, glycolaldehyde, glyceraldehyde, glycerol and 1,3-dihydroxyacetone.

These entries form a connected family of ring, polyol, carbonyl and hydroxymethyl environments. ORACLE assembles α-D-glucopyranose from D-xylose `aconf10` and glycerol `C5`: the placements share four heavy atoms, cover all 12 target heavy atoms and give a maximum overlap RMSD of 0.33 Å. On the rounded values shown, this corresponds to approximately 24–30 times the 0.011–0.014 Å range for helicene and approaches the declared 0.35 Å boundary. Thus helicene probes near-rigid overlap, whereas glucose probes flexible placement; both recover the intended ring and a valid persistent state. Library composition and complete records are given in the Supporting Information.

A leave-one-chemical-identity-out experiment tests reconstruction from an independent donor by withholding 1,3-dihydroxyacetone `c,c` and excludes any donor with the same constitution. The target SMILES and conformer label are fixed before construction, and the hidden high-level geometry is reserved for scoring. ORACLE selects an erythrulose subgraph and reconstructs the specified cis/cis hydroxyl arrangement with heavy- and all-atom RMSDs of 0.067 and 0.144 Å, respectively; the bond-length root-mean-square (RMS) error is 14.0 mÅ. A deterministic 20-start Experimental-Torsion Knowledge Distance Geometry, version 3, calculation followed by the Merck force field provides a fragment-free baseline and gives 0.077 and 0.157 Å and 13.1 mÅ. The comparable skeleton and bond errors isolate the added chemical information supplied by the fragment route: in this case it primarily improves hydroxyl orientation. The reconstructed hydroxyl torsions remain cis at $+16.3^\circ$ and $+17.7^\circ$, while the reference values are 0°; ORACLE accordingly assigns C_1, while the reference is C_{2v}. This predeclared, held-out conformer provides a quantitative test of reconstruction and stereochemical auditing through the residual torsional distortion and associated symmetry change.

Frozen peptide population synthesis. Gly–Ala–Gly extends the same idea to electronic populations while keeping the geometry frozen. The reference is a completely extended, zero-contact configuration, so $s_c(\mathbf{R}_{\mathrm{ref}}) = 0$ and the assembled CM5 populations directly define $\mathbf{q}^0$.

Any subsequently accepted hydrogen bond adds donor polarization, acceptor polarization and charge transfer through its continuous dimer rule. The same parser generates the stereochemical Cartesian structure; ORACLE assigns residue cores, expands each by two covalent shells and caps the outer halo. Charges and Mayer orders are taken from the deepest owned interiors, while repeated halo observations measure boundary sensitivity.

Three L0 fragment calculations and an independent calculation of the complete frozen molecule provide a direct audit (Figure 14). The assembled CM5 charges differ from the direct result by 0.003 e root-mean-square and 0.010 e at most, while restoring exact molecular neutrality requires a maximum correction of 3.90×10^{-4} e. The core-owned construction avoids averaging boundary and interior populations alike. and interior populations alike. Assigning ownership to the core removes cap contamination and preserves both geometry and electronic level.

Neutral fast-multipole source groups are constructed only after the molecular CM5 and Mayer state has been assembled and its total charge restored by the minimum-displacement projection. The QM fragments themselves are therefore overlapping electronic environments; the neutral electrostatic groups represent the assembled molecular state. Separating these two partitions preserves local chemistry during electronic reconstruction and subsequently suppresses the monopole term of neutral groups during far-field evaluation.

Here the service layer supplies stereochemical parsing, matching, valence completion and deterministic hydrogen placement, while GFN-FF provides the universal initial relaxation, chosen for periodic-table coverage and improved accuracy over the Universal Force Field.[22,32] ORACLE then freezes topology, symmetry, local descriptors, atom types and primitive coordinates, keeping the chemical state independent of parser, seed generator and force field.

The current 2025 library cache contains 776 local Cartesian files across its three reference-level collections[31] and already supplies aromatic tiles such as benzene, naphthalene, acenaphthylene and acenaphthene. The helicene construction identifies useful additions more precisely than a request for "larger fragments": compact aza-aromatic, hetero-fused, saturated-

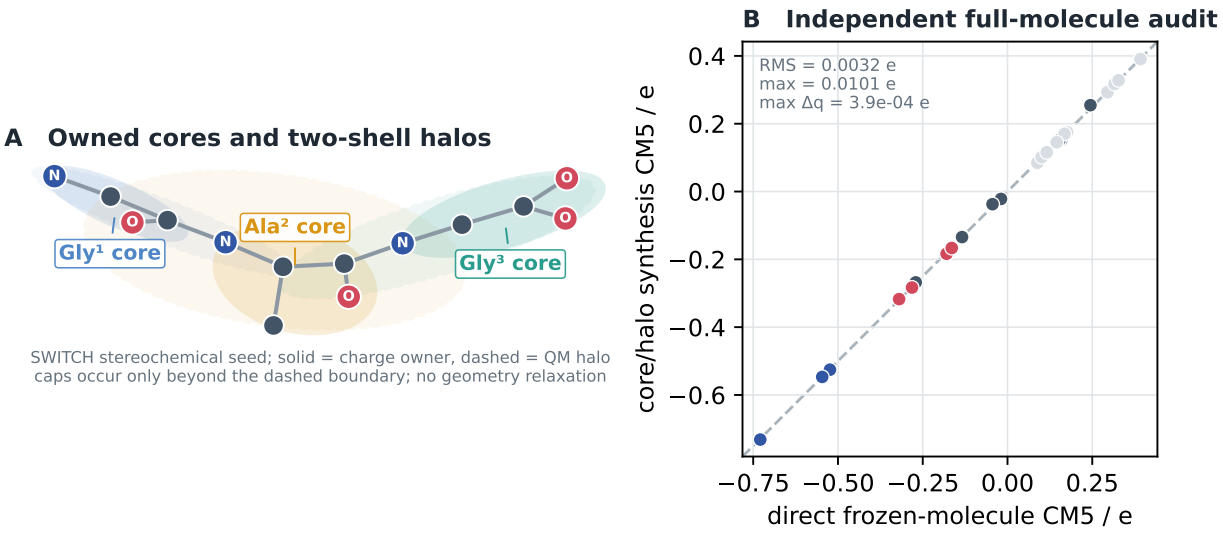


Figure 14: Frozen Gly–Ala–Gly population synthesis from overlapping fragments.

ring, carbonyl-ring and cis/trans junction tiles with non-collinear overlaps. Each record should retain charge/multiplicity provenance and its high-level geometry source.

All uses of this library in the present work—saccharin retrieval, leave-one-out validation, benchmarks and overlap assembly—resolve records through the same manifest-managed local cache. The cache is populated once from the published archives and records their hashes. The reference subset is one deterministic set of 331 usable coupled-cluster singles and doubles with perturbative triples [CCSD(T)] geometries extrapolated to the complete-basis-set (CBS) limit and including core–valence (CV) contributions, denoted CCSD(T)/CBS+CV throughout the work.[31] This state-of-the-art structural level is the source of the library's accuracy; ORACLE separately audits whether its chemical coverage supports a query. The scope of the library differs from that of the formulation. Continuous perception, coordination, local descriptors, and the L0 electronic interface are defined across the periodic table, whereas the published LCB25 accurate structure library is deliberately restricted to the elements essential for the chemistry of life: H, C, N, O, F, P, S, and Cl. Accordingly, every LCB25 retrieval, transfer, and reconstruction test reported here uses these elements. ORACLE perception and the fixed-geometry L0 examples retain their periodic-table scope.

6.6 Test case II: topology at molecular scale

Four demanding structures test perception directly from element labels and archived Cartesian coordinates. ORACLE constructs their bonds, rings and atom selections internally (Figure 15).

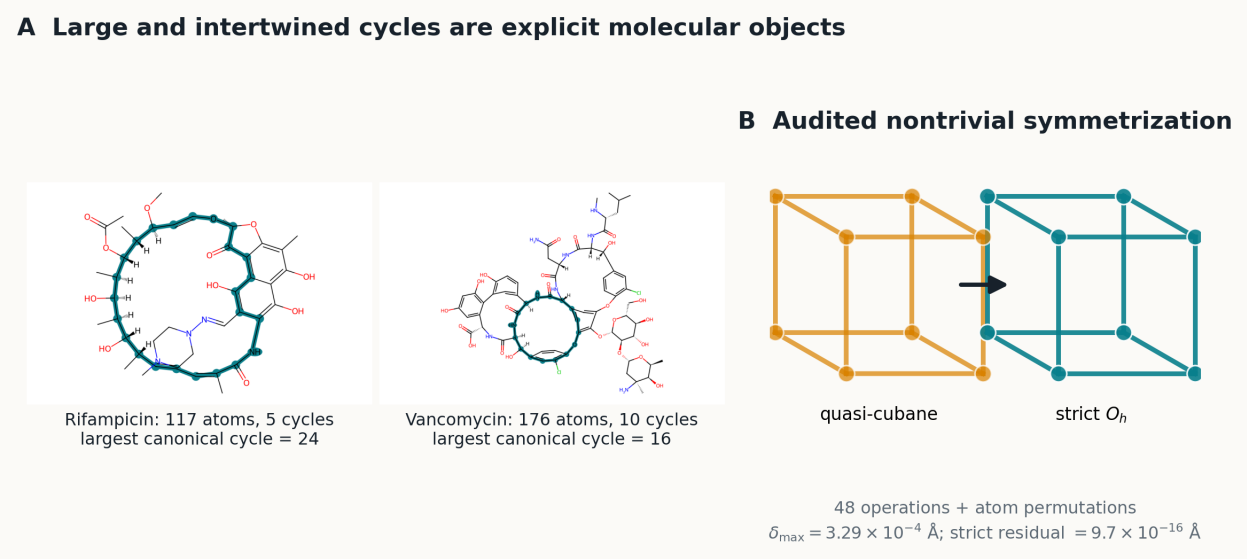


Figure 15: Unrestricted macrocycle perception and nontrivial 48-operation O_h symmetrization.

All four cases recover a complete cycle basis and a Wilson rank of $3N - 6$ (Figure 15). Rifampicin requires an ordered 24-membered ansa cycle within a five-cycle system, while vancomycin requires ten canonical cycles, including two distinct 16-membered loops, in a 176-atom graph. In each case the output is more than a ring count: it is an executable ordered definition consumed by incidence, automorphism and downstream coordinate constructors.

The second panel makes the parallel symmetry claim concrete. Two vertices of an analytic cubane were independently displaced before perception. At the declared 10^{-3} Å tolerance, ORACLE recovers all 48 matrices and atom permutations of O_h; group averaging moves any atom by at most 3.29×10^{-4} Å, and strict reanalysis leaves a 9.7×10^{-16} Å maximum residual. The verified finite action therefore controls the geometry-changing symmetrization.

Synthetic isolated cycles containing 16, 32, 64, 128, 256 and 512 vertices each yield one complete rank-one cycle space. This control demonstrates that the production method retains the complete cycle independently of ring size.

6.7 Test case III: hydrogen bonds add interaction resolution

The water dimer is deliberately simple because it isolates a conceptual point. From Cartesian coordinates alone, ORACLE identifies the covalent components {1,2,3} and {4,5,6}, then perceives the interfragment O1–H2$\cdots$O4 contact. The 1.84 Å H$\cdots$O distance, 180° angle and smooth weight 1.000 are stored separately from the covalent edge set. The same record generates an H2–O4 interaction primitive coordinate.

The same molecular state therefore contains both the two covalent fragments and the hydrogen-bonded assembly as coexisting resolutions. For the calibrated O–H$\cdots$O(H) class, a local model consumes the stored interaction coordinate and produces a +0.020 Å target. This numerical response is included to show how the persistent interaction object is reused: perception establishes which interaction exists, whereas the application model decides what quantity is calculated from it. The reference domain and complete parameters of that model are documented in the Supporting Information.

The same contract was then tested on intramolecular contacts of increasing complexity (Figure 16). Glycine I and II respectively give zero and one accepted hydrogen bond, while retaining one and two weaker interaction closures. The C5 and C7 conformers of N-acetylglycinamide—the glycine-dipeptide analogue—likewise distinguish a weak C5 closure from the accepted N–H$\cdots$O contact of C7.[33] The carbohydrate cases extend the test to oxygen-rich environments. The symmetric *c,c* dihydroxyacetone conformer presents two donors to one acceptor; two LCB26 α-D-xylose conformers reverse the direction of their dominant O–H$\cdots$O contact; and two D-fructose conformers contain three and four accepted contacts within networks of nine and seven interaction closures. For these compact systems, the CM5 reference can already contain one or more intramolecular hydrogen bonds. ORACLE therefore subtracts the tabulated $s_c(\mathbf{R}_{\mathrm{ref}})\Delta\mathbf{q}_c$ contributions on the atoms involved before storing $\mathbf{q}^0$; all later charge changes are generated from this hydrogen-bond-free vector. Thus the peptide and sugar examples exercise the zero-contact and inverse-reference branches of the same equations. Every case retains a full-rank primitive set, while the indi-

vidual contact weights remain continuous—including a marginal 0.075 contact in the more crowded fructose network—whereas acceptance, graph identity and primitive rank are audited explicitly. The result is stable chemical resolution throughout the accepted connectivity basin.

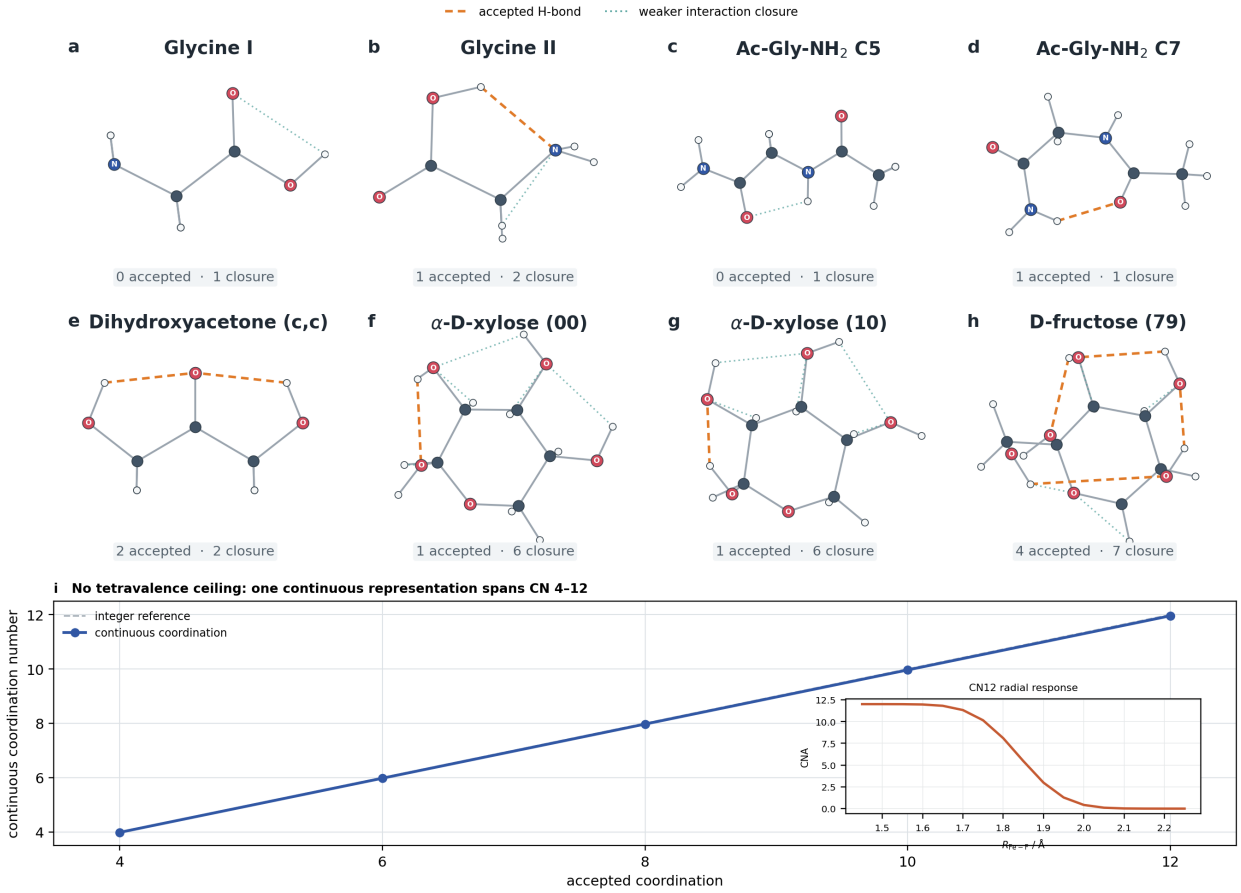


Figure 16: Continuous interaction evidence across peptide-like and carbohydrate motifs, together with coordination numbers 4–12.

6.8 Test case IV: saccharin from perception to coverage diagnostics

Saccharin couples a fused heterocycle, conjugation, polar sulfur bonds and spectroscopic data in a compact 17-atom system. Its geometry-only electronegativity/Pauling observables correlate with the available Complete Charge Model 5 charges and Mayer bond orders:

Pearson/Spearman correlations are 0.862/0.891 for charge and 0.907/0.924 for bond order. Starting from the equilibrium geometry, ORACLE identifies 18 transversal core–valence targets and 13 additional bond targets: six conjugated C–C and four C–H corrections, together with two S–O and one S–N targets examined here. Equation 19 converges in two iterations with a maximum residual of 8.6×10^{-13}.

The existing model leaves residuals of 3.148 and 3.471 mÅ for the S–O and S–N environments. The coordinate reconstruction remains successful, while the coverage record distinguishes these sulfur environments from those already represented. The Supporting Information reports the sulfur diagnostic and Hessian-threshold audit. The corresponding structural audit is shown in Figure 17.

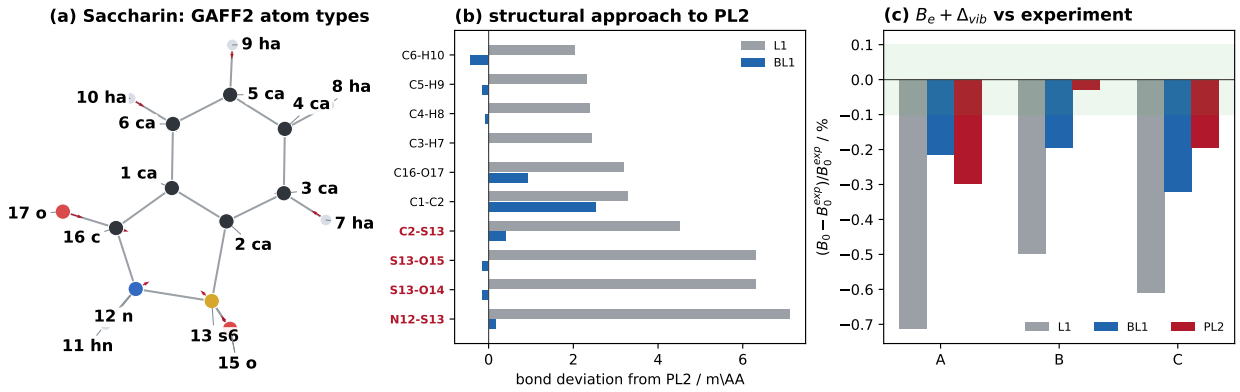


Figure 17: Saccharin structural audit with ORACLE-generated GAFF2 atom types.

Transfer from a compact high-level library is precisely where a chemically informed continuous representation should be most useful. ORACLE perceives all 331 structures at that reference level in the 2025 subset using the same geometry-only information as the saccharin query, resolving whole-molecule similarity into query-directed atom, bond and cycle coverage (Figure 18).

The leading whole-molecule donor, 1H-quinolin-4-one, has combined similarity 0.629. Adding complementary donors raises weighted coverage to 0.808 (Figure 18); sulfur dioxide is selected early because it supplies the best local S–O match despite being a poor global analogue. The best S–N and C–S similarities remain approximately 0.001 or smaller. The

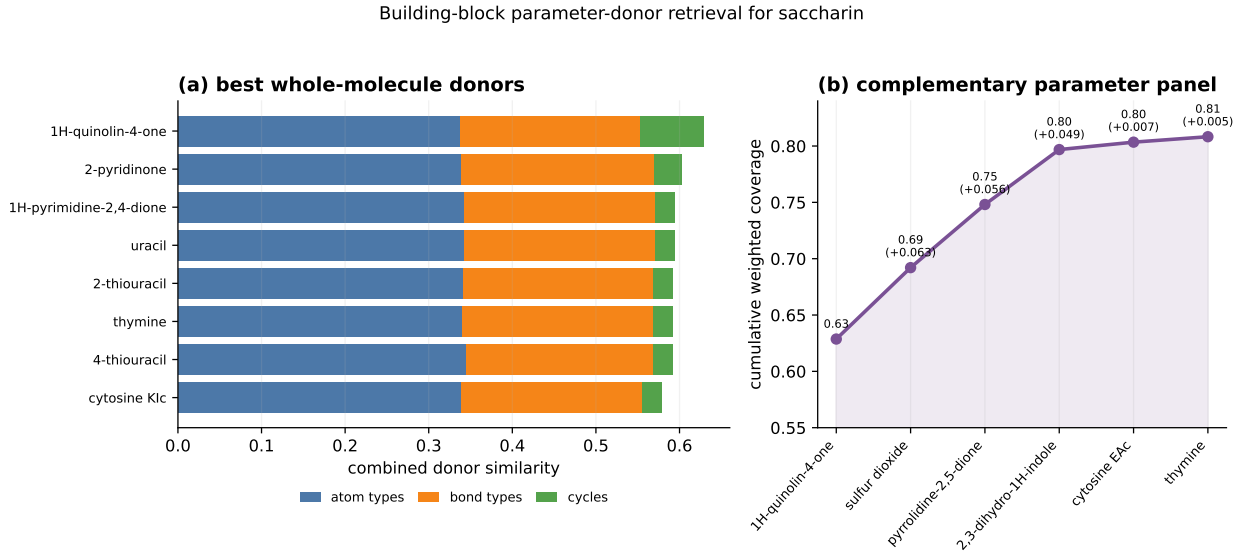


Figure 18: Coverage-aware saccharin retrieval exposes both transferable and missing local environments.

resulting coverage map identifies the distinct joint sulfur-centered block and quantitatively rationalizes the inclusion of saccharin as the first LCB26 member.

6.9 Test case V: held-out local transfer

A strict leave-one-out control removes thiazole from the accurate reference pool, perceives its independent lower-level input geometry, retrieves eight bond parameters from other molecules and imposes them simultaneously through the ORACLE primitive-coordinate back-transformation (Figure 19).

The retrieved donors have mean/minimum similarities 0.909/0.770. Back-transformation converges in two iterations, reducing the bond RMS from 14.1 to 3.2 mÅ and the aligned Cartesian RMS from 14.5 to 5.1 mÅ (Figure 19). This predictive sparse-data test shows that continuous perception can identify useful local donors, while exact topology and primitive definitions ensure that transferred parameters act on the intended molecular objects. Seven targets satisfy the predeclared selector, and thiazole is the detailed case shown here. Together these seven leave-one-out reconstructions provide a quantitative validation of local transfer under a frozen rule: donor selection and reconstruction are completed before the accurate

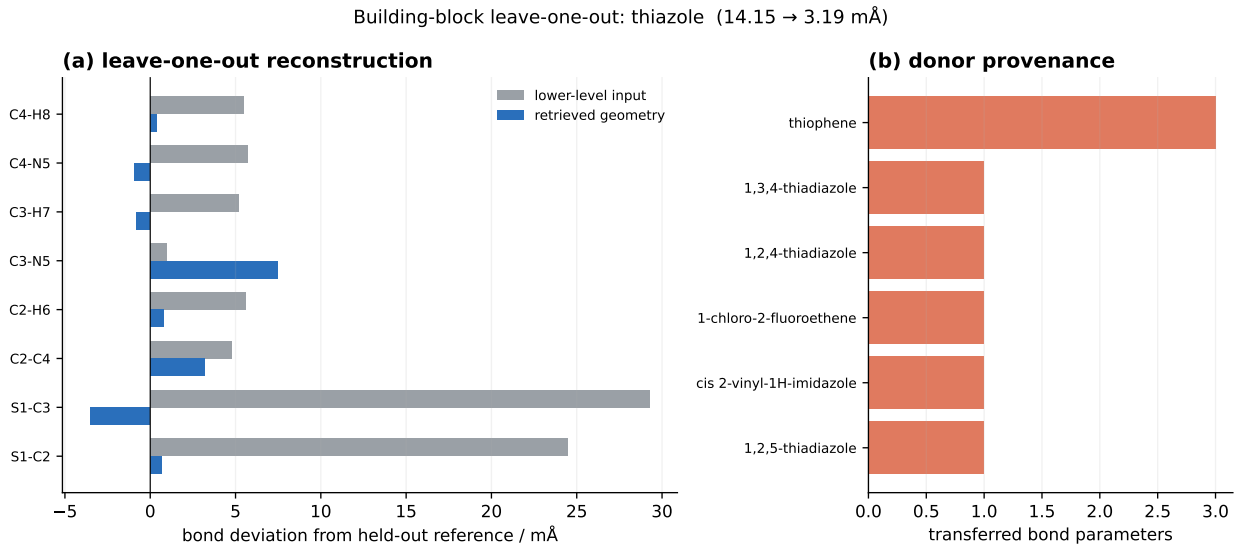


Figure 19: Held-out thiazole reconstruction and provenance of transferred parameters.

geometry of each held-out target is used for scoring.

6.10 Test case VI: from SMILES to a differentially corrected geometry

The more demanding workflow begins from connectivity. Thiazole is parsed and initialized by the same service, relaxed with GFN-FF, and then optimized with the second-generation Geometry, Frequency, Noncovalent extended tight-binding method (GFN2-xTB) before being perceived once. Its accurate reference entry is reserved for final validation. ORACLE identifies 34 required primitives—8 bonds, 11 angles, 12 dihedrals, and 3 out-of-plane coordinates—selects ten distinct library donors, calculates those fragment conformers with GFN2-xTB, and applies Equation 18. Explicit donors support all 34 primitives (Figure 20).

The bond RMS relative to the held-out accurate reference decreases from 13.3 to 4.9 mÅ, the angle RMS from 0.86 to 0.72°, and the aligned Cartesian RMS from 16.1 to 8.6 mÅ (Figure 20). ORACLE applies the 34 transfers simultaneously and then uses the stored six-electron aromatic assignment and C_s action to apply the local-ring and finite-group projectors. Dihedral and out-of-plane RMS deviations remain below 4×10^{-7} degrees. Persistence

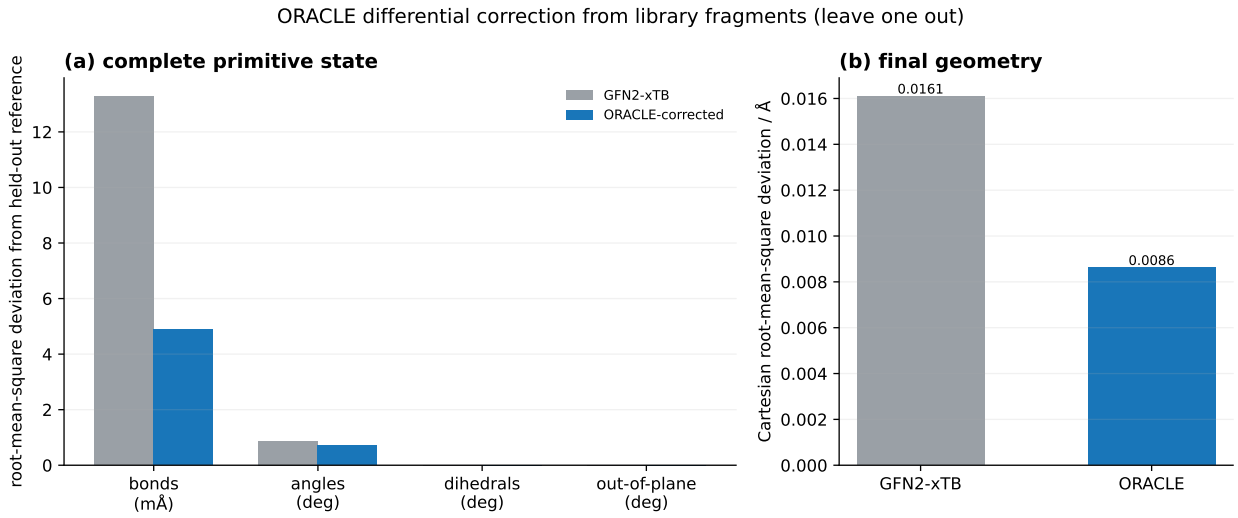


Figure 20: Held-out thiazole after the complete differential workflow.

selects reference information and constrains how that information may modify the molecule.

7 A Common State for Molecular Workflows

The practical implication is a change in workflow semantics. A user no longer passes an unexamined geometry from one application to the next; the user passes a chemically typed state whose identity, evidence and invalidation conditions travel with it. This is particularly important when a medium-to- large molecule is alternately optimized, fragmented, symmetrized, sampled and reconstructed. The state becomes the common scientific object, while files and program calls become transport mechanisms.

For the user, the practical interface is simple: an organic SMILES string or an enriched Cartesian/QCSchema record enters, and a consistent state within the supported chemical domain emerges. That state contains local descriptors and atom types, charges and bond orders, rings, aromaticity, symmetry and fragments, together with constructed or optimized structures when requested. Fragment definition, charge assignment and symmetrization therefore share one graph.

Continuity makes local environments stable under small geometric changes, while exact rings and symmetry preserve the discrete objects required by calculation. Persis-

tence records the thresholds and turns coverage into an actionable quantity: within the H/C/N/O/F/P/S/Cl structural-library domain, saccharin selects a sulfur-centered addition, while the carbohydrate subset documents transfer across oxygen-rich environments.

Fragmentation closes the workflow in both directions. ORACLE decomposes large molecules into overlapping, charge-complete local objects and assembles traceable structures from validated fragments. Every assembled or optimized geometry is re-perceived and audited against the same state.

8 Conclusions

ORACLE establishes a persistent chemical state in which molecular identity is defined by continuous geometric and electronic evidence, accepted chemical objects, and their explicit dependencies. Atomic synthons and the effective atomic number provide the central descriptors: they retain local chemical meaning while remaining suitable for comparison, transfer and parameter construction. This creates a common representation for sophisticated quantum-mechanical workflows on chemically important molecules in the roughly 100–200 atom range.

The defining feature is the bidirectional exchange between perception and quantum mechanics. A single baseline calculation supplies charges, Mayer bond orders and spin information that refine the chemical state; the refined state then supports geometry generation, atom typing, force-field selection, fragment assembly and subsequent electronic-structure calculations. The same mechanism distinguishes spin multiplicities, excited states, solvent environments, transition states and high-coordination topologies at fixed or changing geometries.

The validation demonstrates this scope through held-out structural transfer, overlapping-fragment reconstruction, hydrogen-bond response, open-shell and excited-state populations, reactive coordinate charts and polynuclear coordination. These applications share one per-

sistent and inspectable state, so chemical information remains available as a molecule moves between representations and computational methods. ORACLE therefore provides a general foundation for efficient metrics, intelligent fragmentation, solvent-aware and state-aware force fields, and reproducible integration of advanced quantum chemistry across molecular science.

Data and Code Availability

The data and software underlying this study are supplied in the Supporting Information and in the public versioned repository `https://github.com/yogibubu/ORACLE-manuscript`. The repository contains the source, validation inputs, scripts and machine-readable records used here; the ORACLE source is distributed under the BSD 3-Clause License.

Reproducibility

The Supporting Information reports the complete theoretical definitions, validation tables, input fingerprints, ordered ring sequences and exact reproduction commands. All reproduction scripts use portable relative paths.

Author Contributions

Vincenzo Barone conceived the ORACLE methodology, developed the scientific formulation and software, performed the validation, and wrote the manuscript.

Conflict of Interest

The author declares no competing financial interest.

Supporting Information Available

Supporting Information includes the complete mathematical formulation, parameter and notation definitions, structural-improvement models, topology and aromaticity corpus, exact large-ring definitions, finite-group residuals, quasi-symmetry threshold sweep, descriptor-invariance and atom-class audits, symmetric-cubane projection, baseline charge and bond-order acquisition, overlapping charge synthesis, complete hydrogen-bond response vectors and fixed-geometry metal-complex single points, compiled/Python backend equivalence, saccharin and building-block-library numerical records, serialized-state tests and exact reproduction commands; auxiliary-code versions; Avogadro reimport semantics; and the complete [6]helicene and carbohydrate overlap-assembly manifests, coordinates and tests.

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